
Machine learning and deep learning in evolutionary genetics

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+ credits for some slides and tutorials: J Cury, T Sanchez, A Quelin



Announcements

- **LEGEND conference on machine learning for evolutionary genomics**, 3rd edition should be in Germany in **2027** (see previous edition: <https://legend2025.sciencesconf.org/>)
- In France: LEGO working group <https://gt-lego.cnrs.fr/>

Tenure track position in AI for genomics/evolutionary genomics opening in my lab (LISN, Paris-Saclay) or in LBBE (Lyon), France. **Please advertise in your groups!** (people can contact me at flora.jay@lisn.fr for any question)

<https://www.ins2i.cnrs.fr/en/cnrsinfo/list-cnrs-informatics-tenure-tracks-2026>



Outline

Machine Learning: basic concepts and terminology

ML, application to popgen ; neural networks

- I. From ABC to deep learning for population genetics**
- II. Learning directly from SNP data with neural networks
- III. Dissecting two published networks for effective population size inference
- IV. Opening on applications of unsupervised deep learning to popgen
- V. Hands-on: building/training/re-using ML and DL models with application to population genetics (demography/selection)
 - ML: scikit-learn
 - DL: dnadna <https://mlgenetics.gitlab.io/dnadna/>

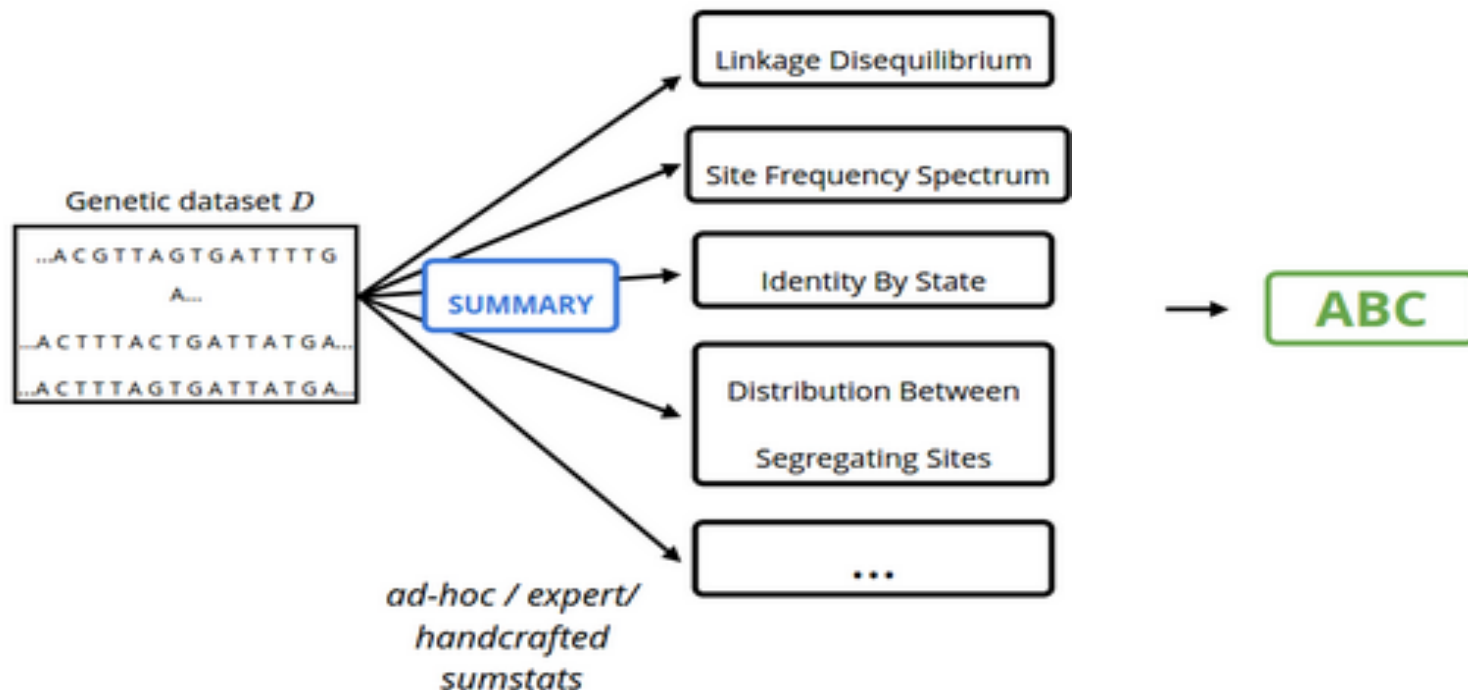
Reviews: Deep Learning in evolution

- Borowiec, Marek L., Rebecca B. Dikow, Paul B. Frandsen, Alexander McKeeken, Gabriele Valentini, and Alexander E. White. “Deep Learning as a Tool for Ecology and Evolution.” *Methods in Ecology and Evolution* 13, no. 8 2022: 1640–60.
<https://doi.org/10.1111/2041-210X.13901>.
- Korfmann, Kevin, Oscar E Gaggiotti, and Matteo Fumagalli. “Deep Learning in Population Genetics.” *Genome Biology and Evolution*, 2023, evad008.
<https://doi.org/10.1093/gbe/evad008>.
- Yelmen, Burak, and Flora Jay. “An Overview of Deep Generative Models in Functional and Evolutionary Genomics” *Annu. Rev. Biomed. Data Sci.* 2023. 6:1–18
<https://doi.org/10.1146/annurev-biodatasci-020722-115651>

Approximate Bayesian Computation:

likelihood free inference based on simulations

- Data summarized by **handcrafted summary statistics**
- **Real and simulated** summary statistics are **compared**
- The comparison informed on the likely demographic scenario
- Application to demography: Boitard et al 2016 , Jay et al 2019 and many other works



Here and in all DL methods presented afterwards training is based on large datasets of simulated data with labels (i.e. for which we know the evolutionary parameters)

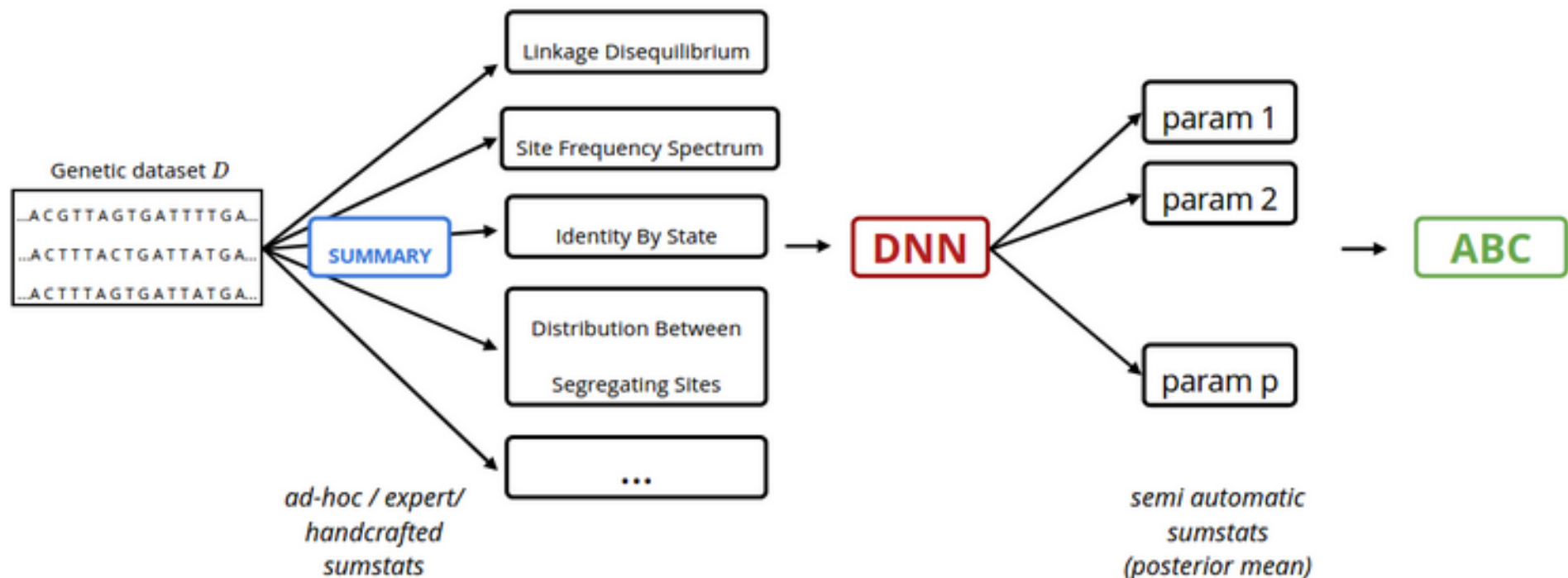
Deep learning on summary statistics (+ABC)

- Generally fully connected net / multilayer perceptron (MLP), e.g.:

- Selection and demo inference, [Sheehan and Song 2016 \(no ABC\)](#)
- Model selection+inference (archaic admixture models), [Mondal et al. 2019 \(with ABC\)](#)

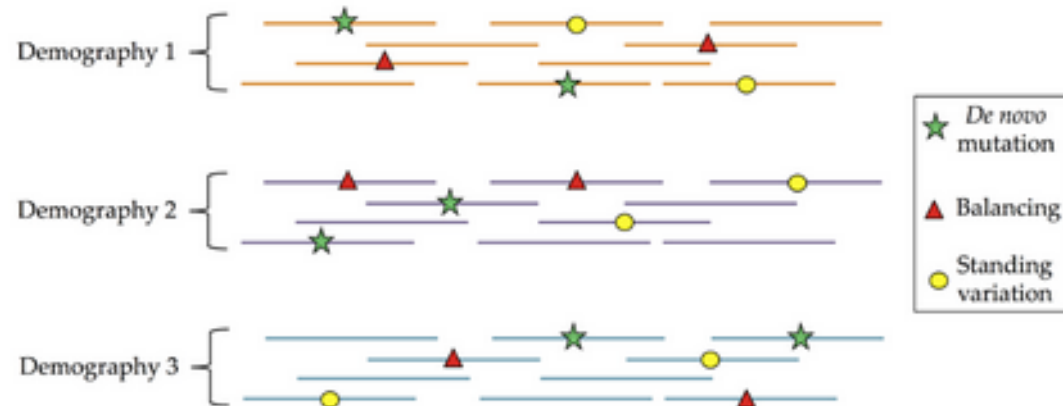
- Those were inspired by Jiang et al 2017 (MLP)

but see as well Creel 2017 (MLP), Raynal et al. 2017 (random forest), Fernhead and Prangle 2012 (posterior mean as $s(\cdot)$)

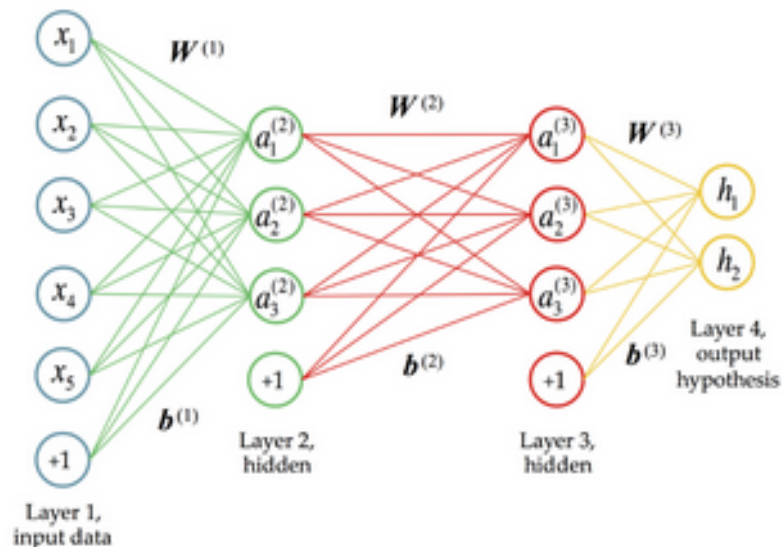


Deep learning on summary statistics

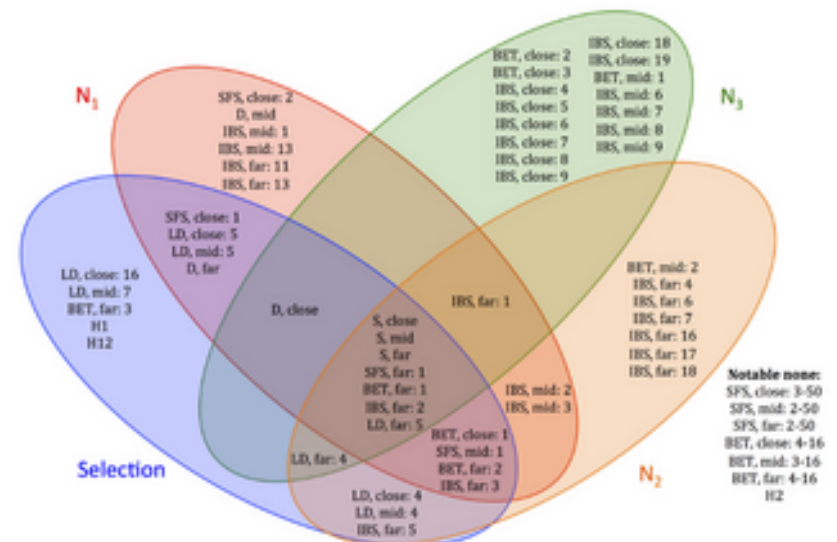
- Generally fully connected net / multilayer perceptron (MLP), e.g.:
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MLP:



Importance of summary statistics based on permutation:



Deep learning on summary statistics

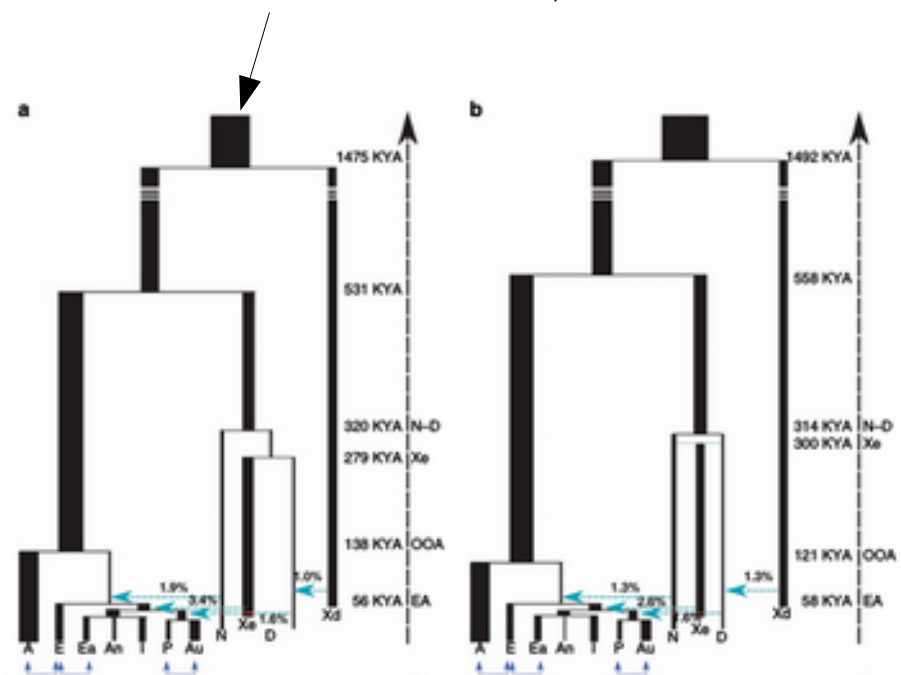
- Generally fully connected net / multilayer perceptron (MLP), e.g.:
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Mondal et al. 2019 : ABC [MLP (joint SFS)]

2 models selected among 8 models + parameter estimation

-> third archaic introgression in Asia and Oceania from Neandertal-Denisova clade or from Denisova related lineage (early divergence)

Ancestors of modern humans, Neandertal and Denisova



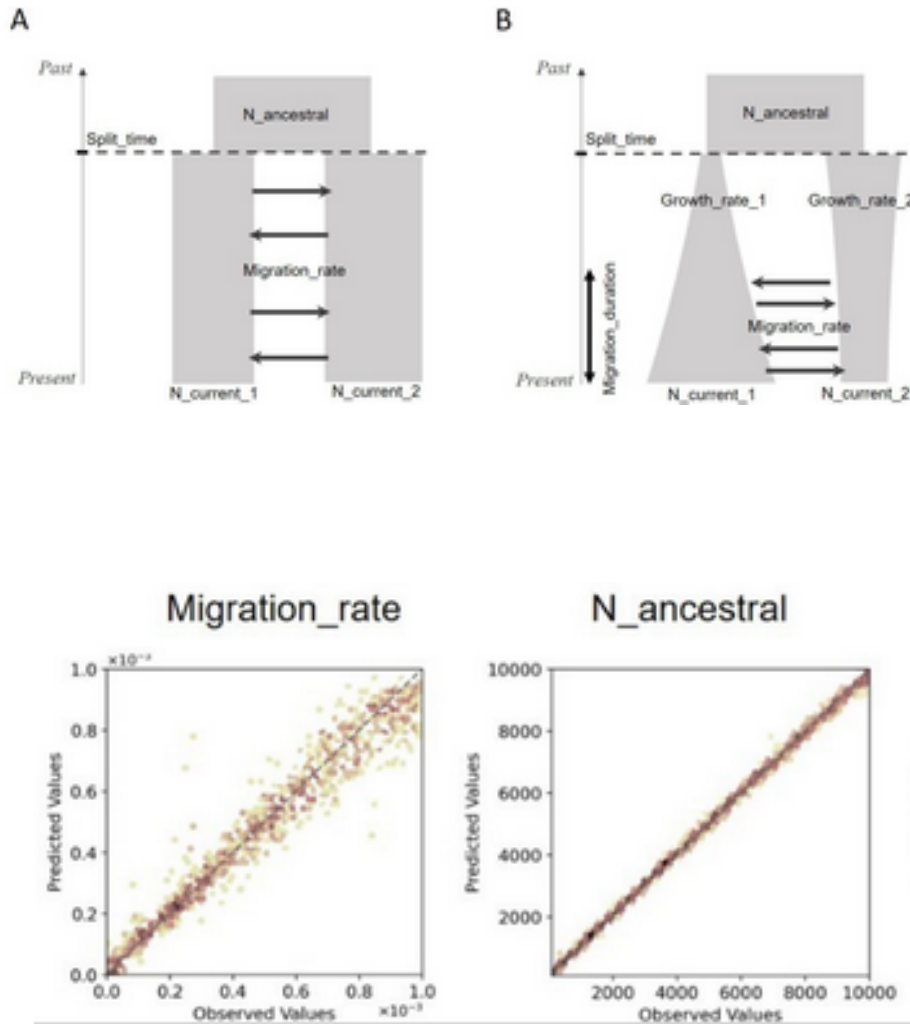
Present-day
humans

Archais
Neandertal+
Denisova

Present-day
humans

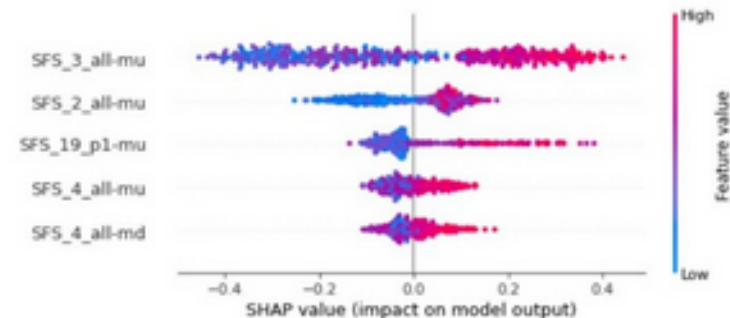
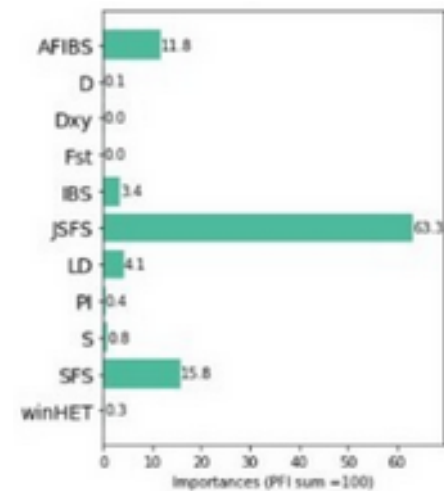
Archais
Neandertal+
Denisova

Deep learning on summary statistics – cf tuto

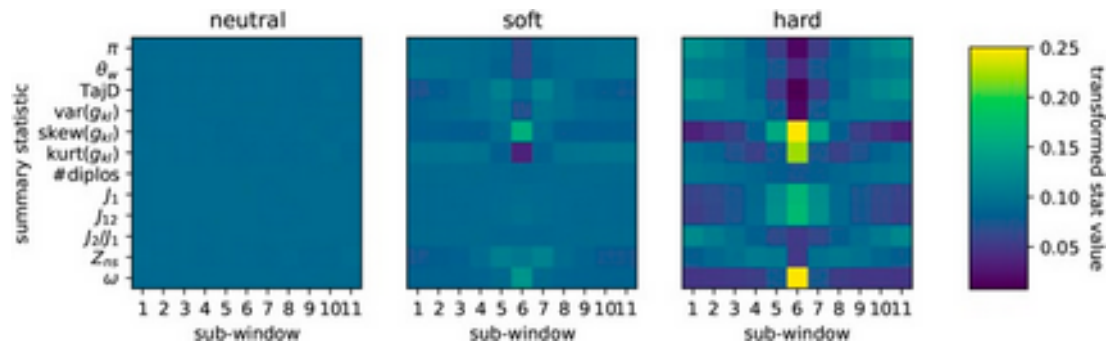


Quelin et al (Heredity 2025)

- Goal: estimate the demographic parameters of two-population models
- Fully connected network (compared to RF, XGBoost and ABC) on SFS, joint SFS, IBS, AF-IBS, Fst, diversity pi, etc.
- Measure of summary statistics importance based on permutations and SHAP values (shown below for Split_time)



Deep learning on summary statistics

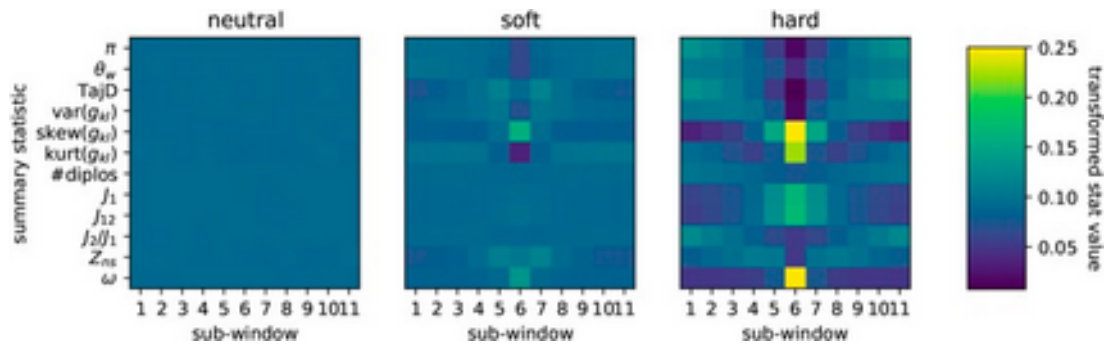


goal: classify selection/neutral regions of the genome

summary statistics along the genome

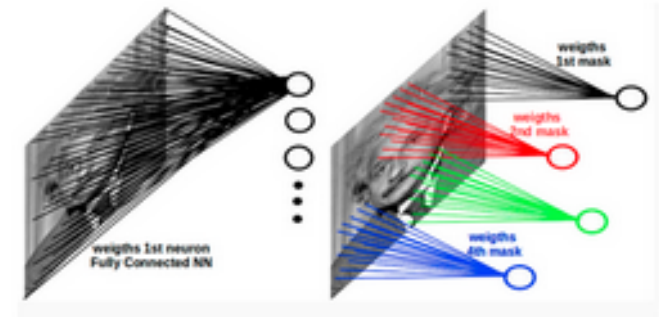
Kern & Schrider MBE 2018
(see follow-up in Xue et al MBE 2021)

Deep learning on summary statistics



Fully connected

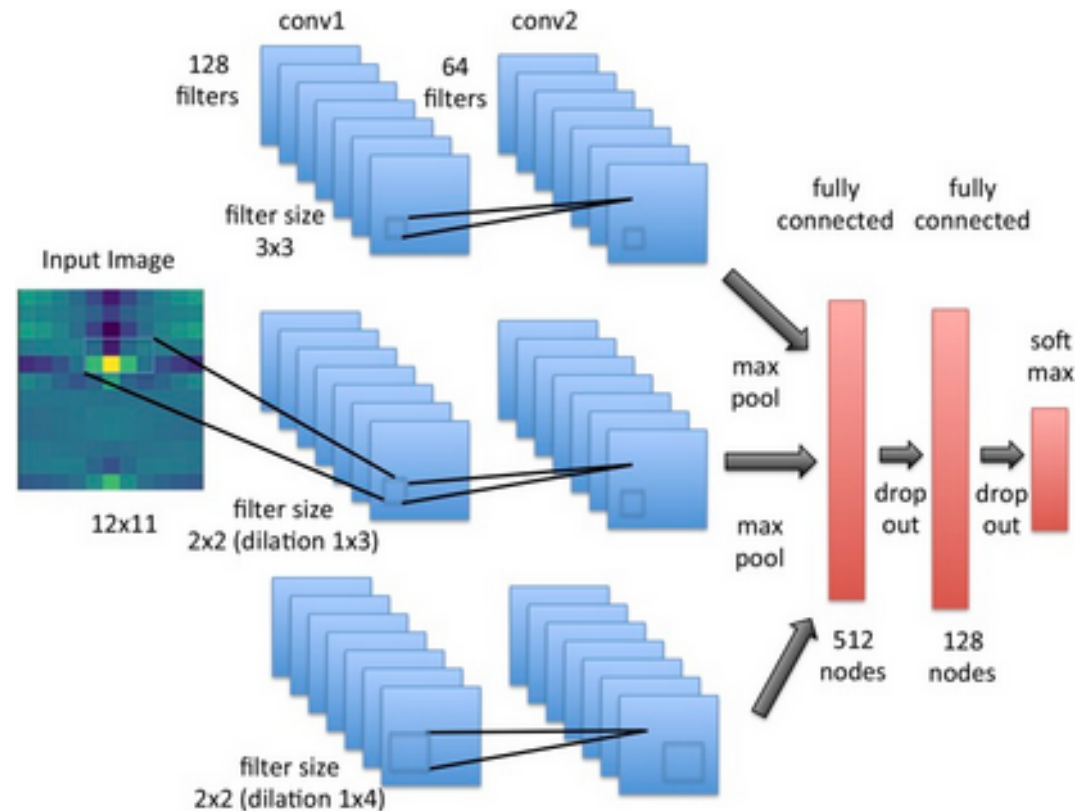
Convolutional



goal: classify selection/neutral regions of the genome

summary statistics along the genome

“images” of sumstats processed by a **Convolutional Neural Network (CNN)**



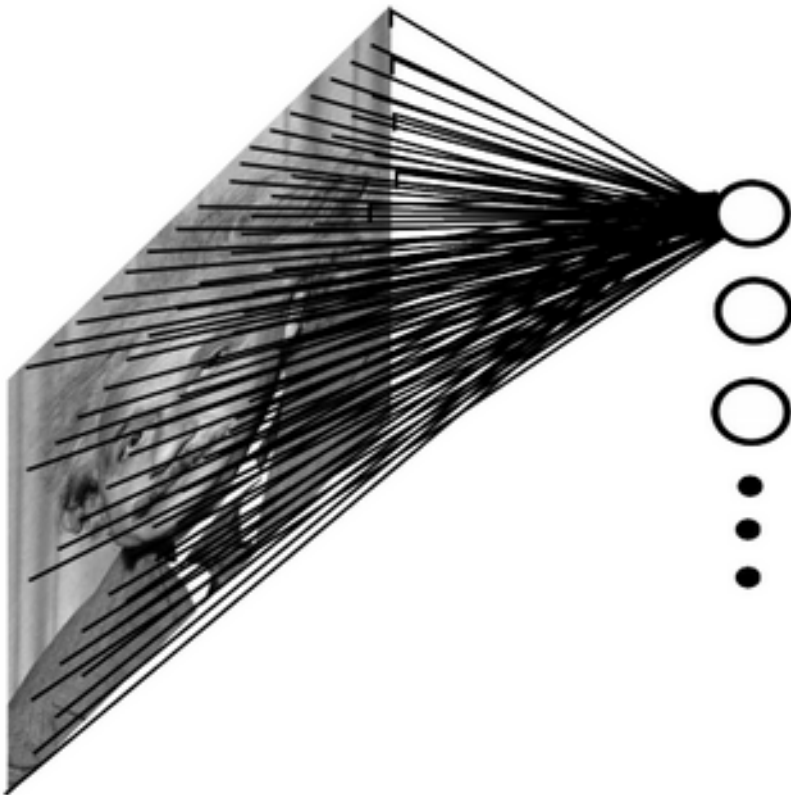
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DL - convolution

- Why using convolution networks? (convnet)

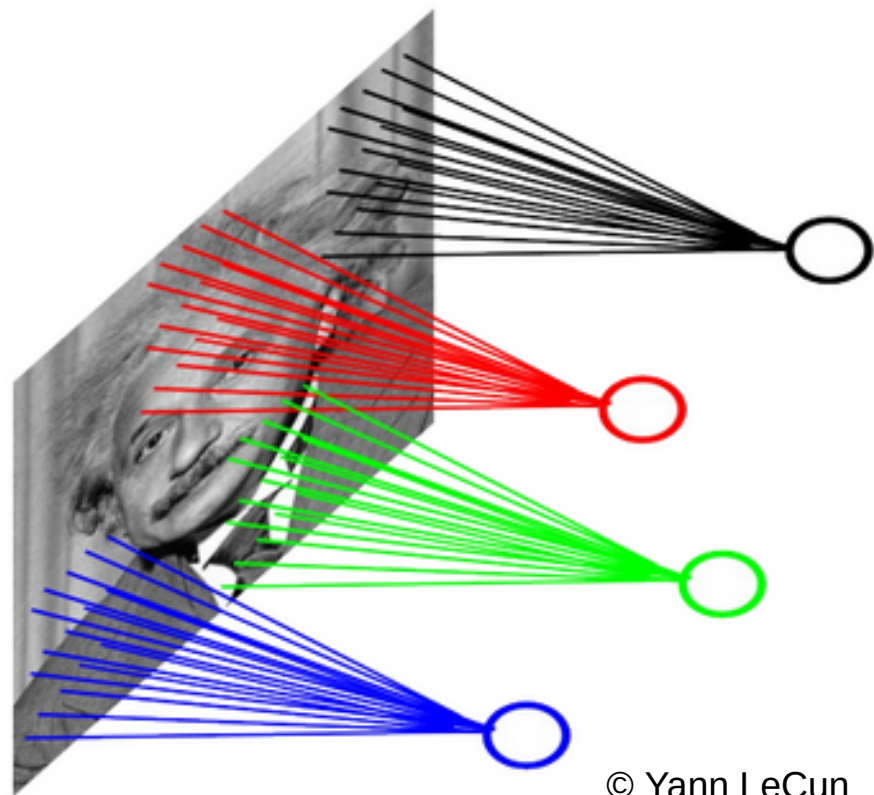
Fully connected :

→ huge number of parameters to learn (each edge correspond to a weight)



Convnet

locally connected
→ smaller number of parameters to learn

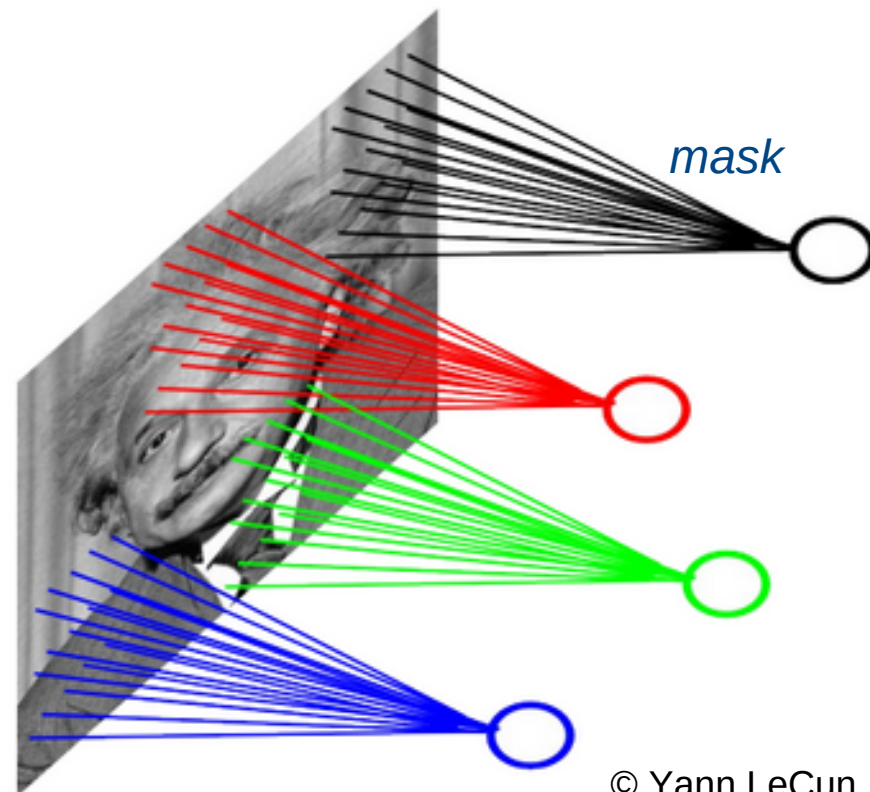
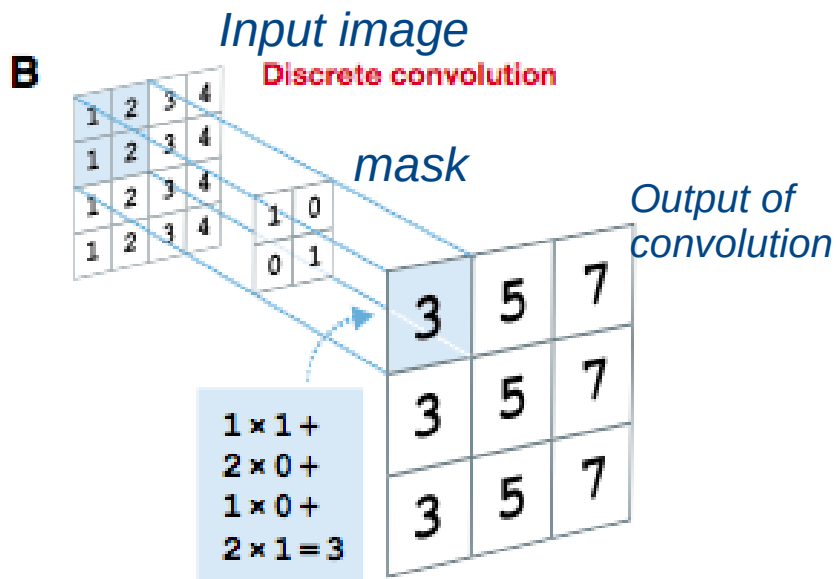


© Yann LeCun

DL - convolution

- Why using convolution networks? (convnet)

Convnet
locally connected
→ smaller number of
parameters to learn



© Yann LeCun

Convolution operation

- Convolution with padding and stride=1 → no dimension reduction

0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

Kernel		
0	-1	0
-1	5	-1
0	-1	0

114	328	-26	470	158
53	266	-61	-30	344
403	116	-47	295	244
108	-135	256	-128	344
314	346			

- No padding and/or stride > 1 → dimension reduction

60	113	56	139	85
73	121	54	84	128
131	99	70	129	127
80	57	115	69	134
104	126	123	95	130

Kernel		
0	-1	0
-1	5	-1
0	-1	0

266	-61	-30
116	-47	295
-135	256	-128

Exercise

- Computer by hand a convolution operation (see below)
- Compute the number of parameters
- Design a 3x3 filter (with fixed weights) that could detect horizontal edges (detect a pattern ~ maximal activation for this pattern)

Input data

```
0 1 1 0 0
1 1 0 1 0
1 0 0 0 1
```

Convolutional
operation (with
fixed **weights**)

0.5	-1	2
-1	0.5	3

Output ?

Exercise

- Computer by hand a convolution operation (see below)
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Input data

0 1 1 0 0
1 1 0 1 0
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Input data

0	1	1	0	0
0	1	0	1	0
1	0	0	0	1

Convolutional
operation (with
fixed weights)

0.5	-1	2
-1	0.5	3

Output ?

1.5 2.5 1

Exercise

- Computer by hand a convolution operation (see below)
- Compute the number of parameters
- Design a 3x3 filter (with fixed weights) that could detect horizontal edges (detect a pattern ~ maximal activation for this pattern)

Input data

0	1	1	0	0
0	1	0	1	0
1	0	0	0	1

Convolutional
operation (with
fixed weights)

0.5	-1	2
-1	0.5	3

Output

1.5	2.5	1
-2	2.5	2

Exercise

- Computer by hand a convolution operation (see below)
- Compute the number of parameters
- Design a 3x3 filter (with fixed weights) that could detect horizontal edges (detect a pattern ~ maximal activation for this pattern)

Input data

0	1	1	0	0
0	1	0	1	0
1	0	0	0	1

Convolutional
operation (with
fixed weights)

0.5	-1	2
-1	0.5	3

Output

1.5	2.5	1
-2	2.5	2

Output of max
pool operation

2.5

Exercise

- Computer by hand a convolution operation (see below)
- Compute the number of parameters
- Design a 3x3 filter (with fixed weights) that could detect horizontal edges (detect a pattern ~ maximal activation for this pattern)

Input data

```
0 1 1 0 0
0 1 0 1 0
1 0 0 0 1
```

Convolutional
operation (with
fixed **weights**)

```
0.5  -1  2
-1   0.5  3
```

Output

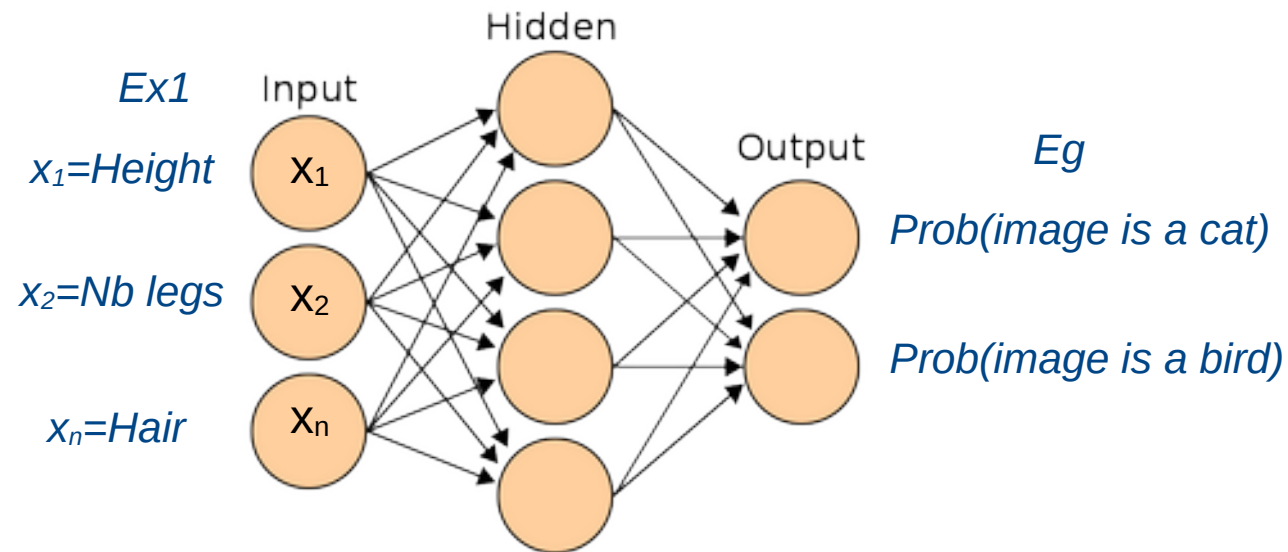
```
1.5  2.5  1
-2   2.5  2
```

OR applying a
RELU activation
and another conv
filter (e.g. 2x2)

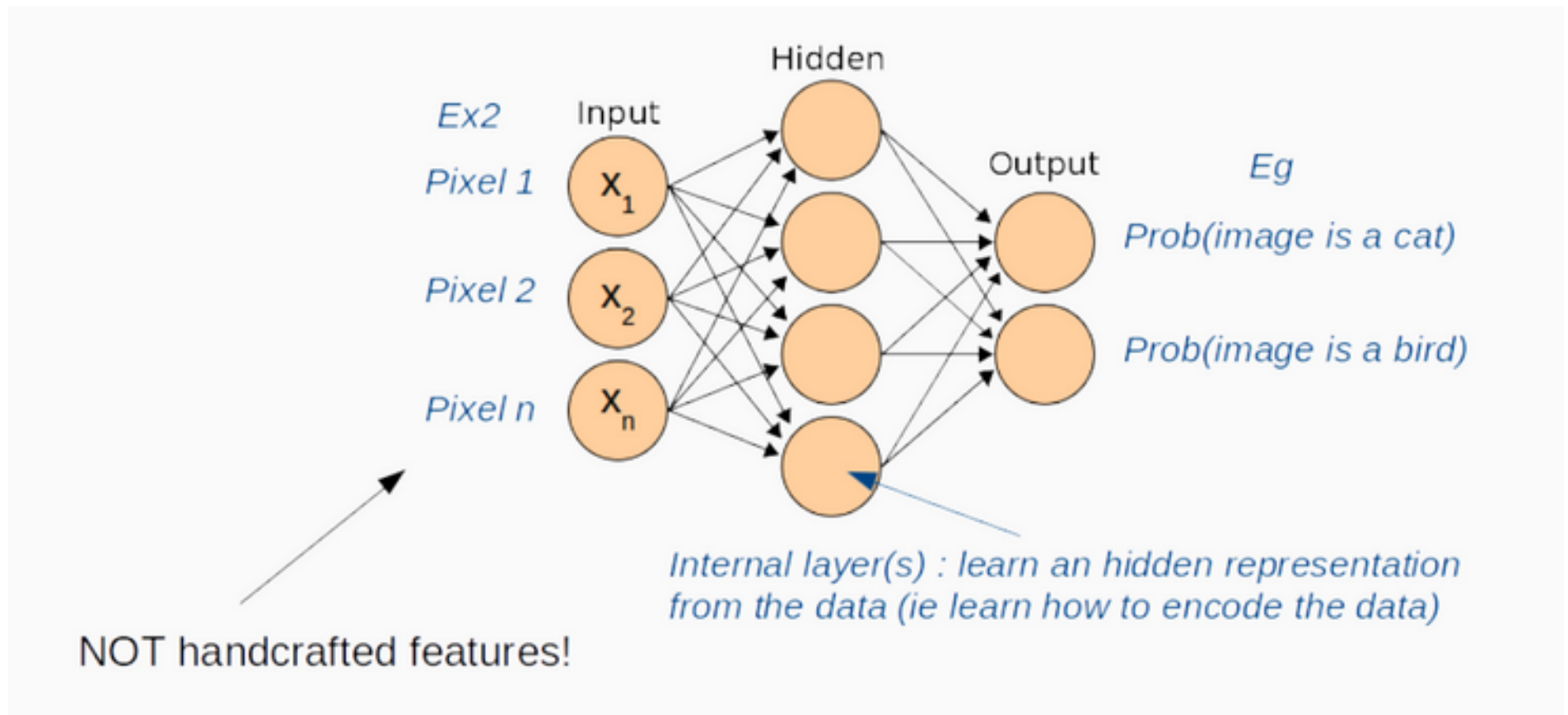
```
1.5  2.5  1
0   2.5  2
```

Reminder: you could bypass summary statistics

From summary statistics (handcrafted features):



Reminder: you could bypass summary statistics



Outline

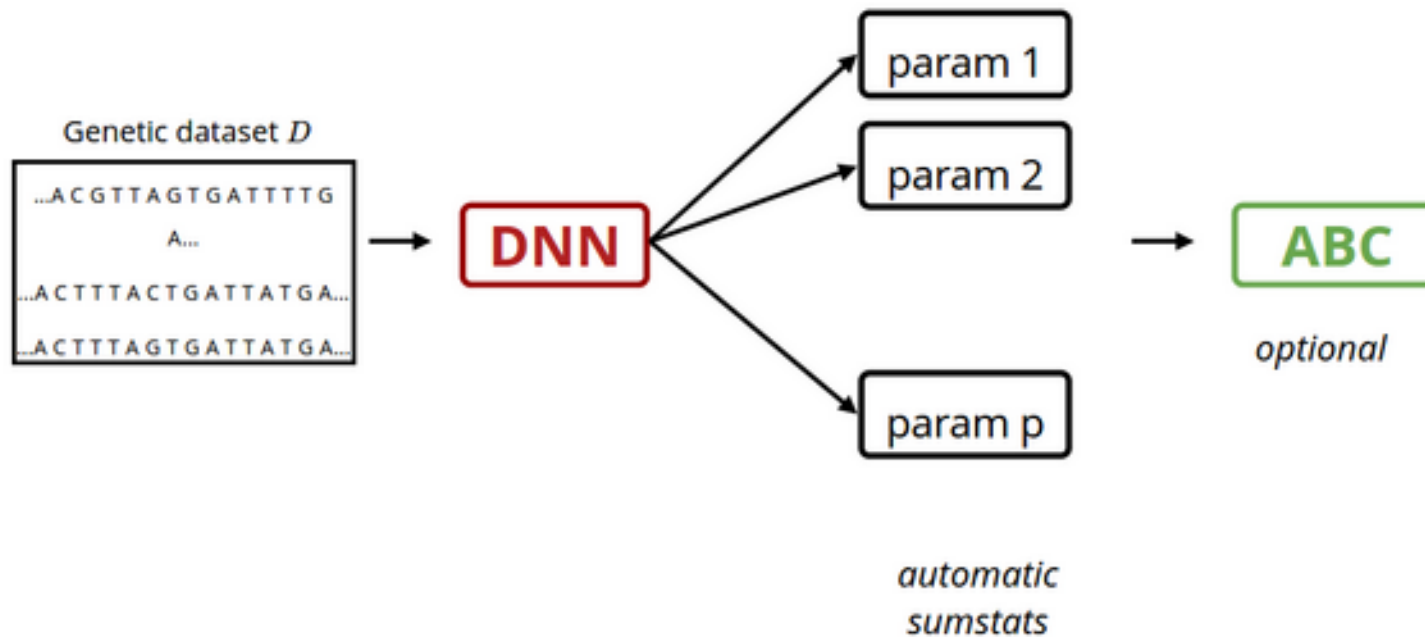
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Deep learning on "raw" genetic data

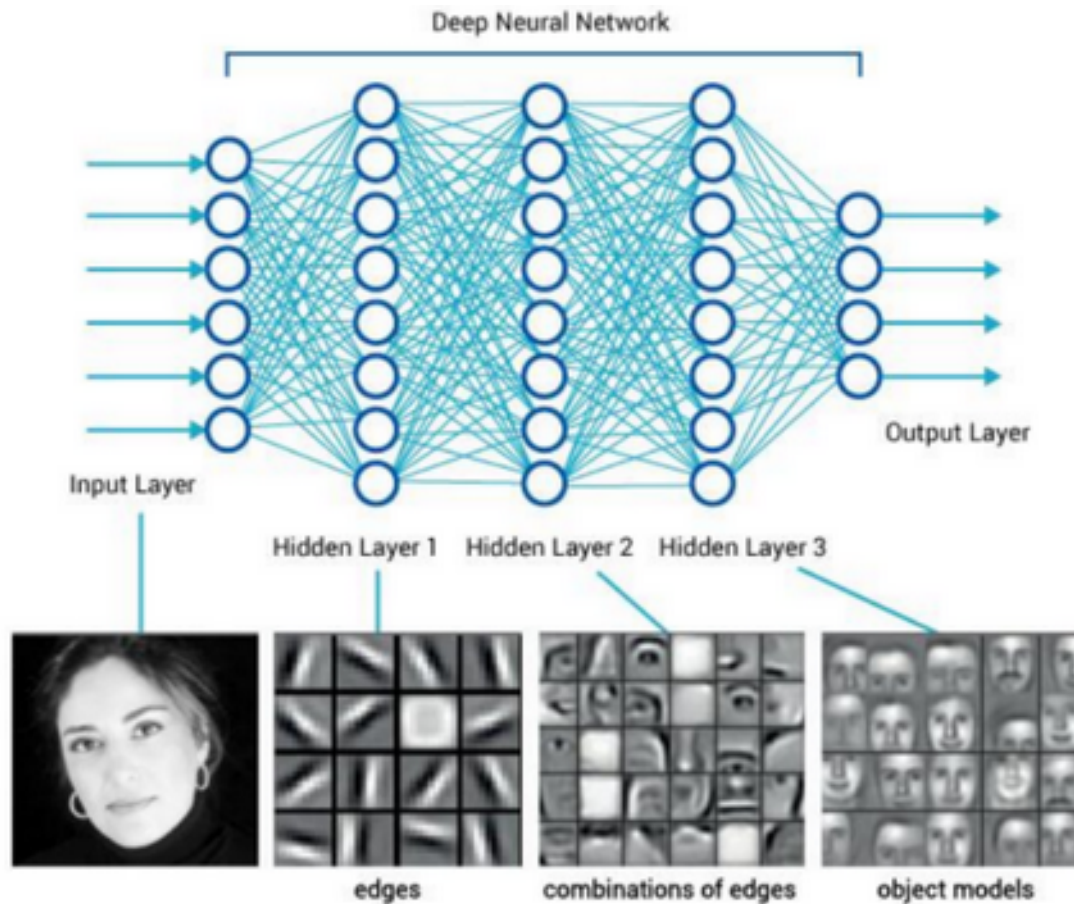
- Process directly the genetic data to bypass handcrafted features
- Often convolution neural networks (CNN)
- Inspired by Jiang et al 2017 but previous works in popgen skip the ABC step



(e.g. DNN tries to predict $N_1, \dots, N_p$ etc) and these are used later as automatic summary statistics (or automatic features) processed by ABC

DL - learning hierarchical representations

Deep Learning (DL) = deep neural networks = nnet with multiple layers



DL - learning hierarchical representations

- Able to learn a hierarchy of representations with increasing level of abstraction

Eg. for image :

pixel → edge → motif → part → full object → combination (eg landscape, scene)

Eg. for text :

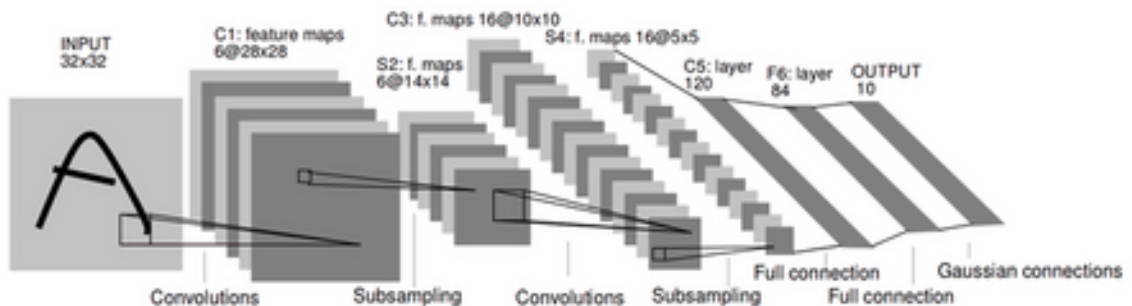
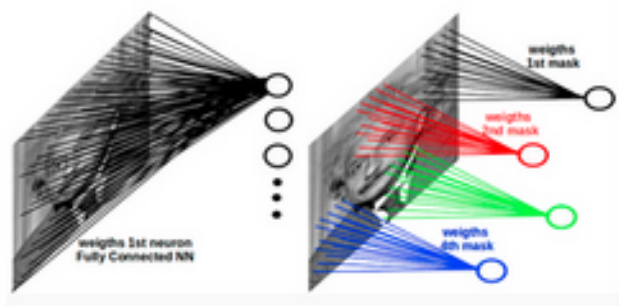
letter → word → word group → sentence → story

...

- A layer = trainable function that transforms input into features at a certain hierarchy level

Deep learning on "raw" genetic data

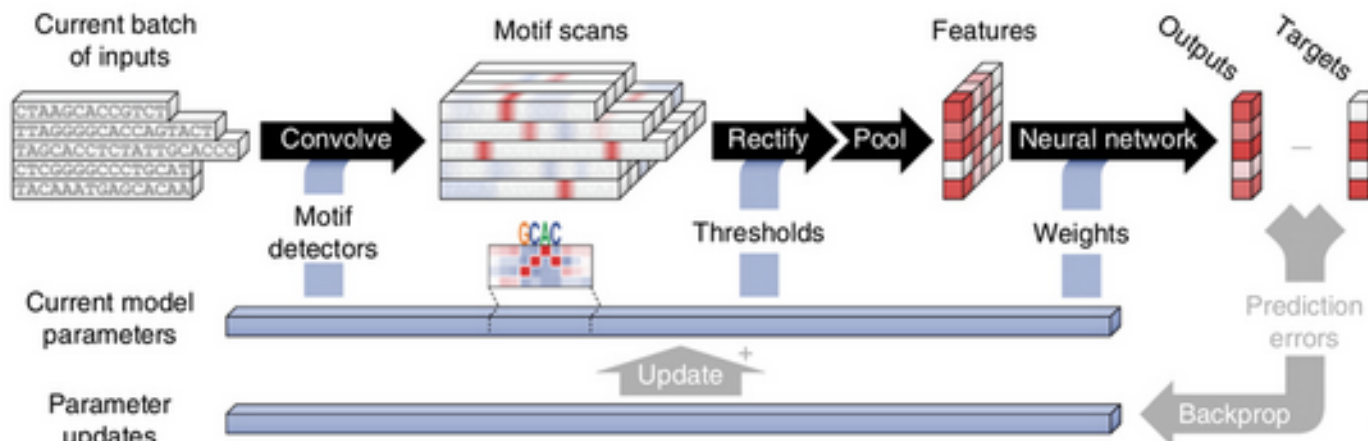
- Convolution networks work well for **computer vision**:



Lecun et al.

1998

- Already used on **DNA sequences**:



Alipanahi et al.

2015

Reminder of steps for a simulation-based supervised ML approaches

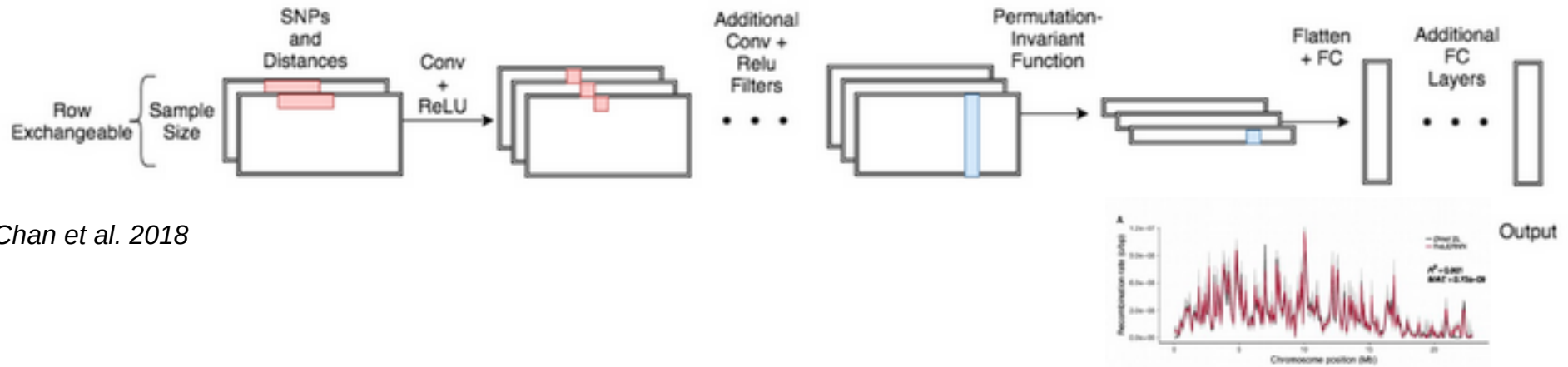
- Define clearly your task
 - Regression/classification of xXx ; score function, loss
 - Define one or several models e.g. Constant size, Fluctuating size, Fluctuating+selection
 - Pick priors for the parameters of these models
e.g. $N_e \sim U[0,100]$, selection coeff $\sim N(0,10)$, ...
- Randomly draw parameters
- Simulate thousands/millions of such SNP matrices thanks to genetic simulators (msprime, msms, slim, bactSLiMulator, ...) using the random parameters
- **Design**, train and evaluate a **model** directly on these matrices

Typical input for population genetics methods



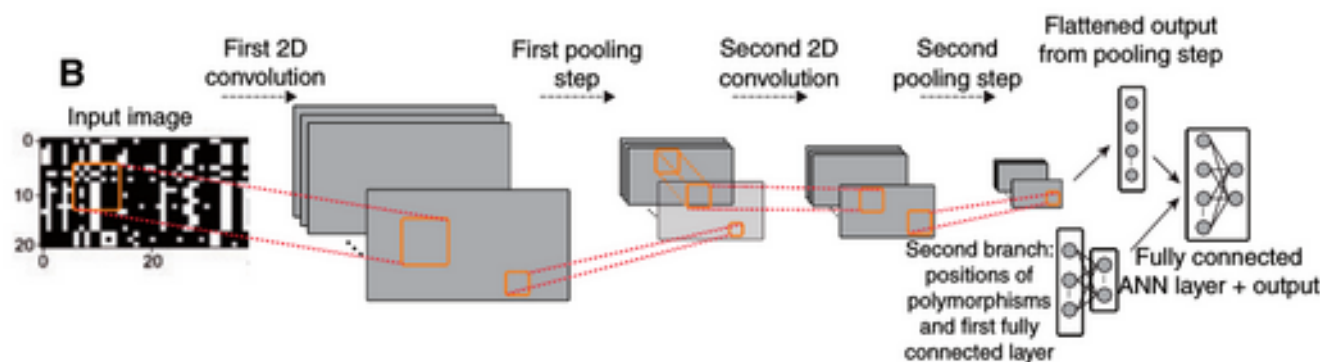
DNN on "raw" data in popgen: SNPs (and distance) matrix

- Detection of recombination hotspot, *exchangeable CNN net*

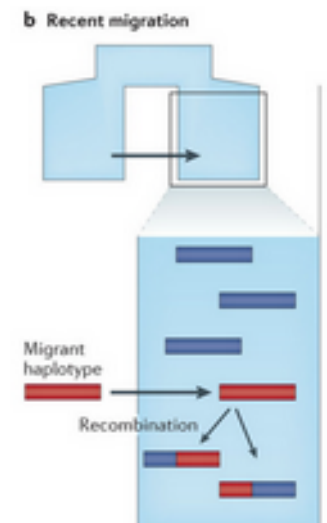


Chan et al. 2018

- Inference of introgression, selection, recombination rate and population size histories with 5 parameters (3-step history), CNNs

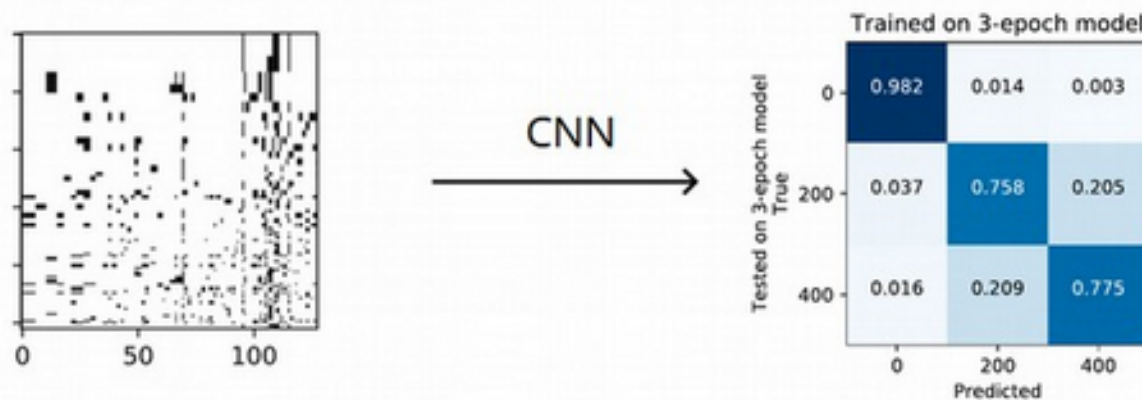


Flagel et al. 2019



DNN on "raw" data in popgen: SNPs (and distance) matrix

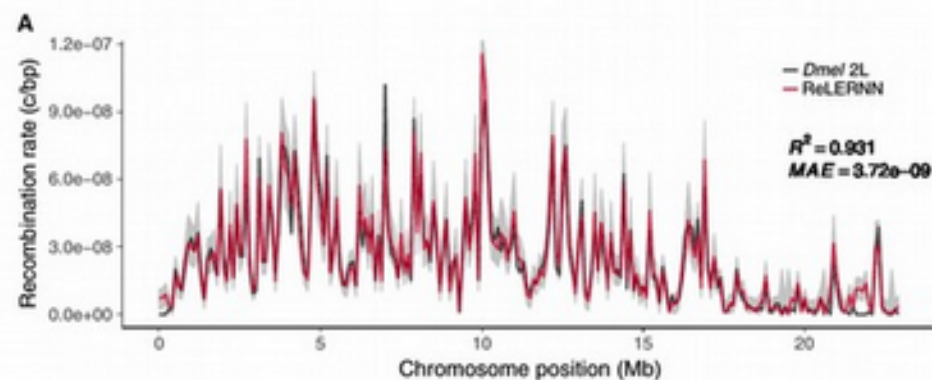
- Predicting selection (under fixed demography), *CNN*



Torada et al.
2019

related: Isildak et al bioRxiv (balancing selection vs incomplete sweep),

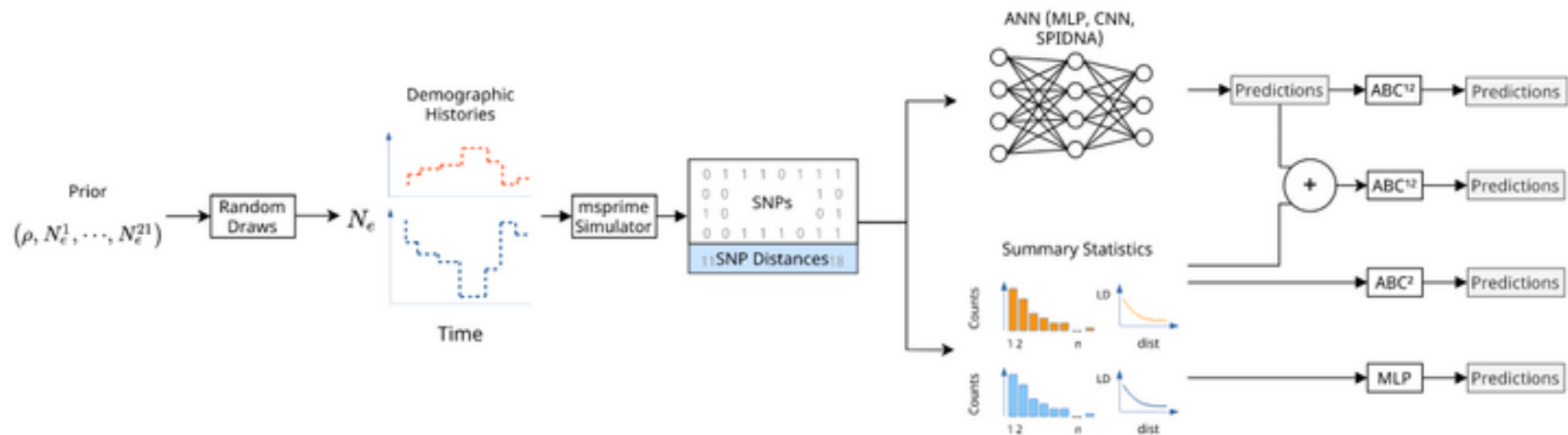
- Inference of recombination with *recurrent networks (RNN)*



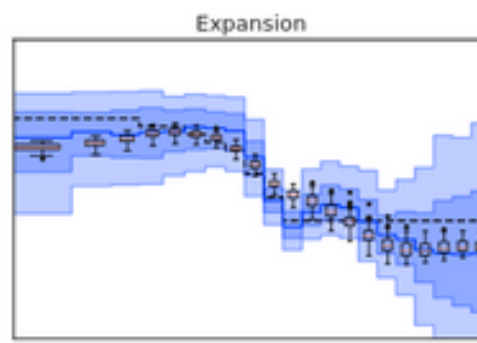
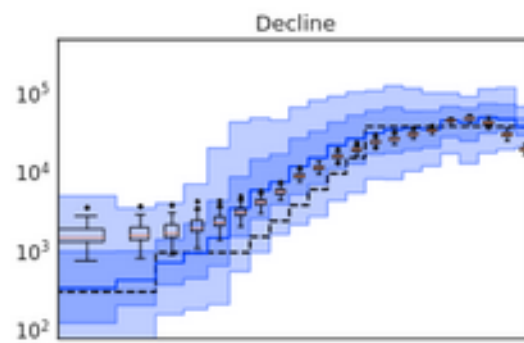
Adrion et al.
2020

DNN on "raw" data in popgen: SNPs (and distance) matrix

- Predicting fluctuating population size (21 steps), exchangeable CNN
- Comparison and combination with ABC



Effective population size (log scale)



Sanchez et al.
2020

DNNs on raw data

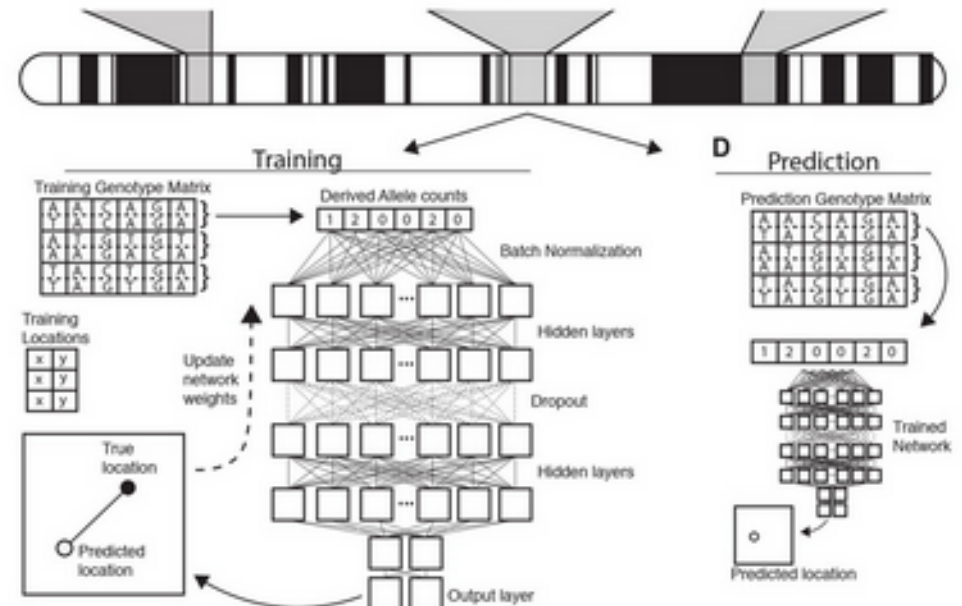
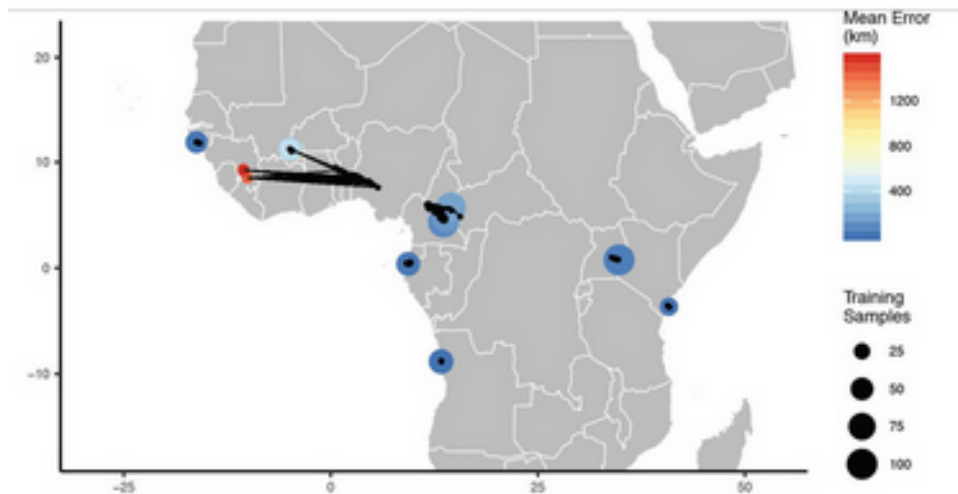
Battey et al 2020

Predicting geographic location from unphased genotype windows

Fully connected network

Trained on “true” labels (geo loc)

Applied to mosquitoes



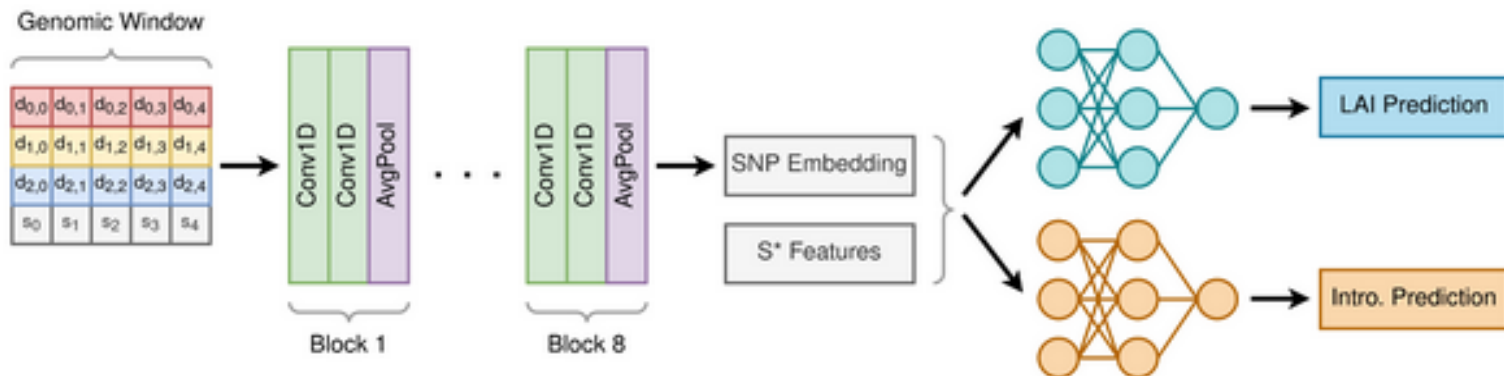
DNNs on raw data+handcrafted features – ancestry inference

Medina-Tretmanis et al 2026a,b

a. Benchmark of Local Ancestry Inference (LAI) DNN-based methods (MLP, CNN, Transformers) + improvement based on handcrafted features (flagging private variants)

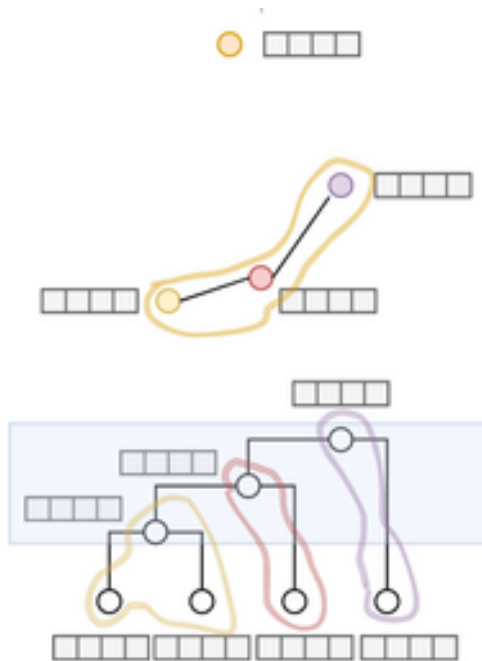
<https://www.biorxiv.org/content/10.64898/2026.03.11.711082v1.abstract>

b. Joint archaic and continental ancestry inference – can be trained on a mix of simulations and real data (preprint in July)



NN exploiting trees for inference

- Estimating **effective population sizes** and **selection** from Ancestral Recombination Graphs using graph convolutions [hierarchical compression i.e. feature computation coherent with the tree]
Korfmann et al PCJ 2024



- Estimating cultural processes from tree imbalance statistics (*Guez PhD thesis 2023*)
- One issue of these approaches is that you need the trees:
 1. estimate them for real data using a software
 2. if this is accurate enough, compare directly to the known simulated trees (training set)
 3. if not, also estimate all trees for the training set using same software as in 1.

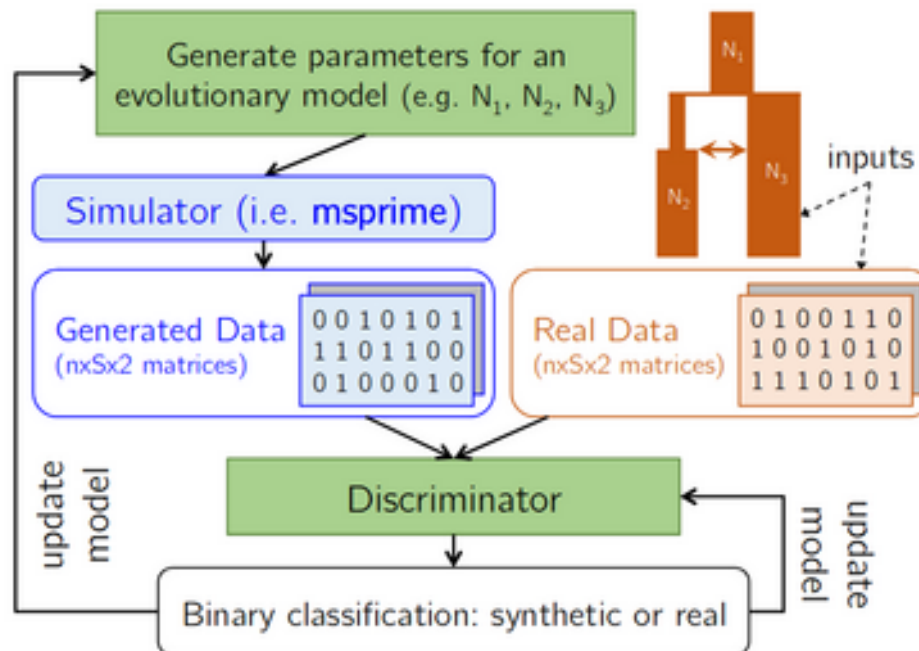
For applications on DNN from tree/phylogenies outside of popgen see epidemiological (Voznica et al 2022) and diversification analyses (Lambert et al Systematic Biology 2023)

Also see the upstream question of predicting species delimitation models (Perez et al MER 2021) or phylogenies (Suvorov et al 2019 ; Zou et al MBE 2020 ; re-evaluation by Zaharias et al 2022 ; Nesterenko et al MBE 2024) from genetic data.

DNNs on raw data

Wang et al 2021. inspired by Generative Adversarial Networks (concerning GAN for genomics see also Yelmen et al PLoS Genet 2021)

- Goal: predicting demographic models (isolation-migration type) from SNP data
- Coalescent **simulator** has to generate realistic SNP matrices
- **Discriminator** trained to distinguish simulated from real
- Discriminator serves as feedback to the coalescent simulator which **parameters are updated via simulated annealing** with the aim to fool the discriminator

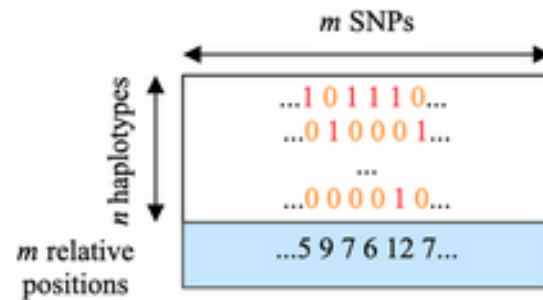


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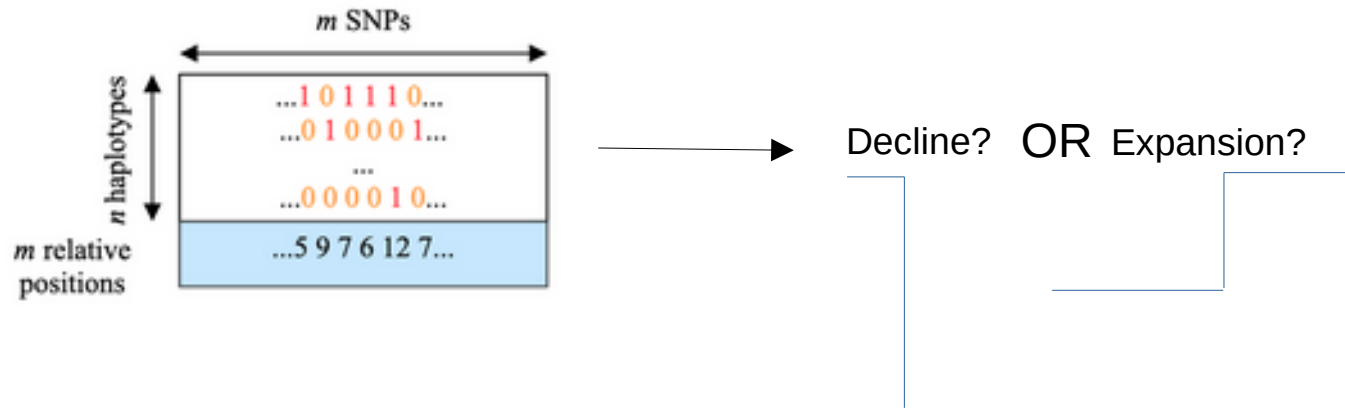
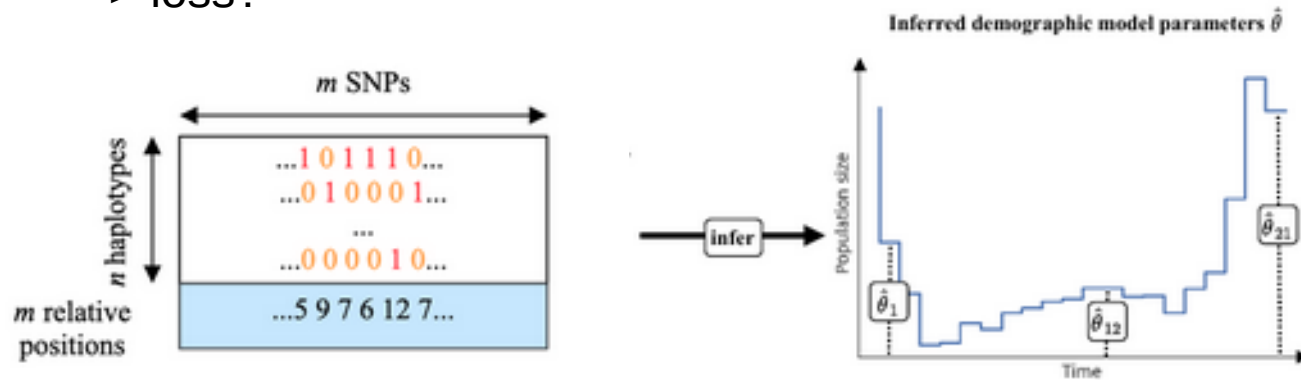


Exercise: define the task and loss for a net that could inform you whether there was a strong decline of effective population size

What's my model(s)? What are the parameters? Regression or classification task? --> loss?

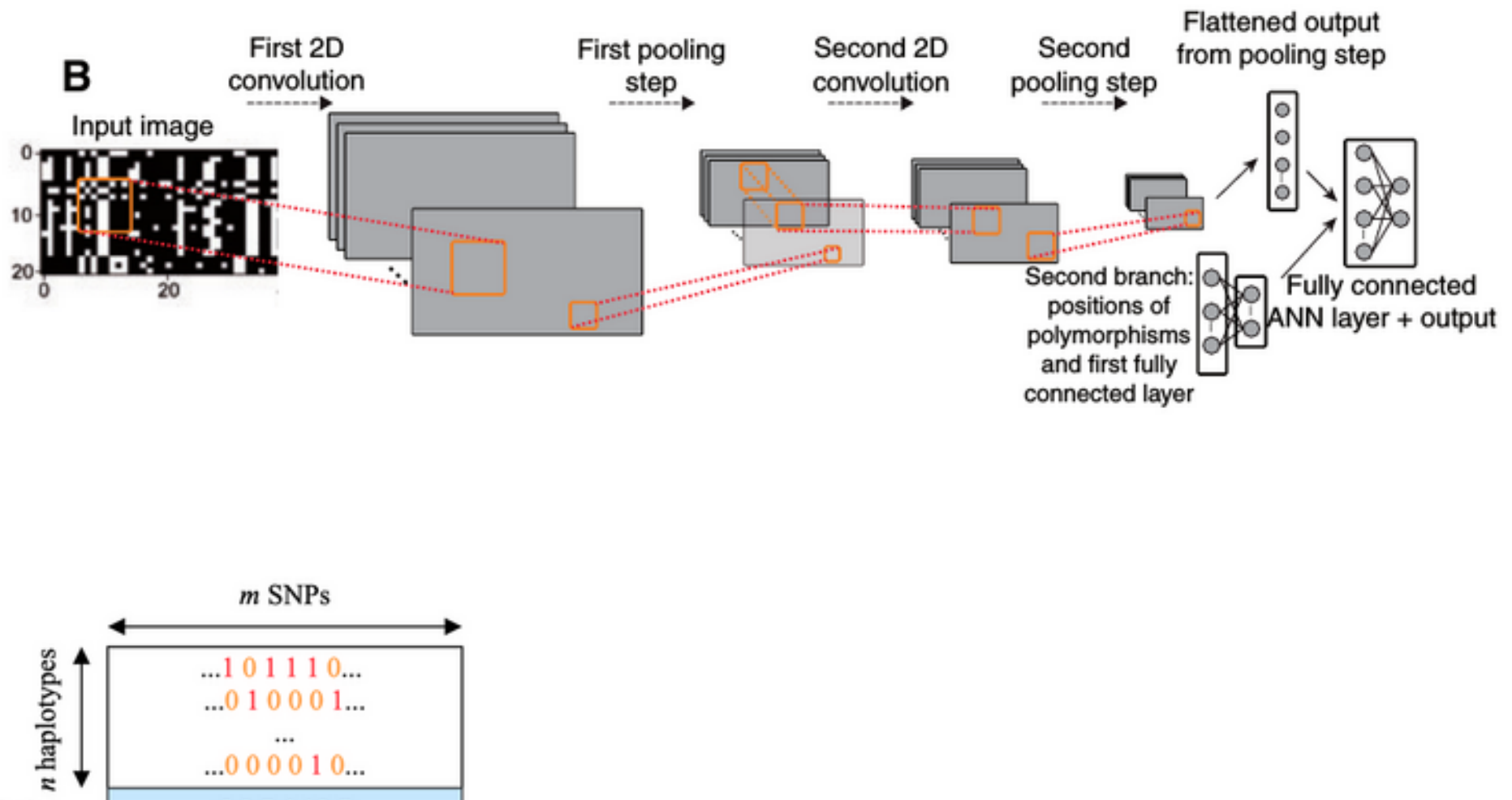
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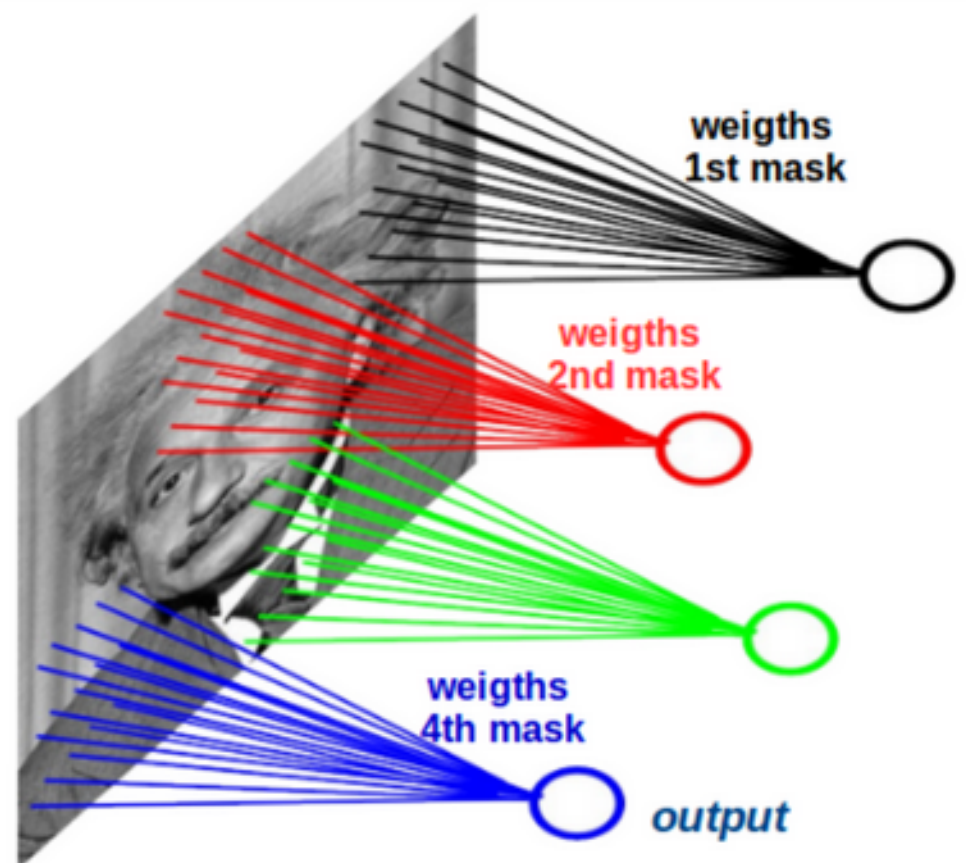
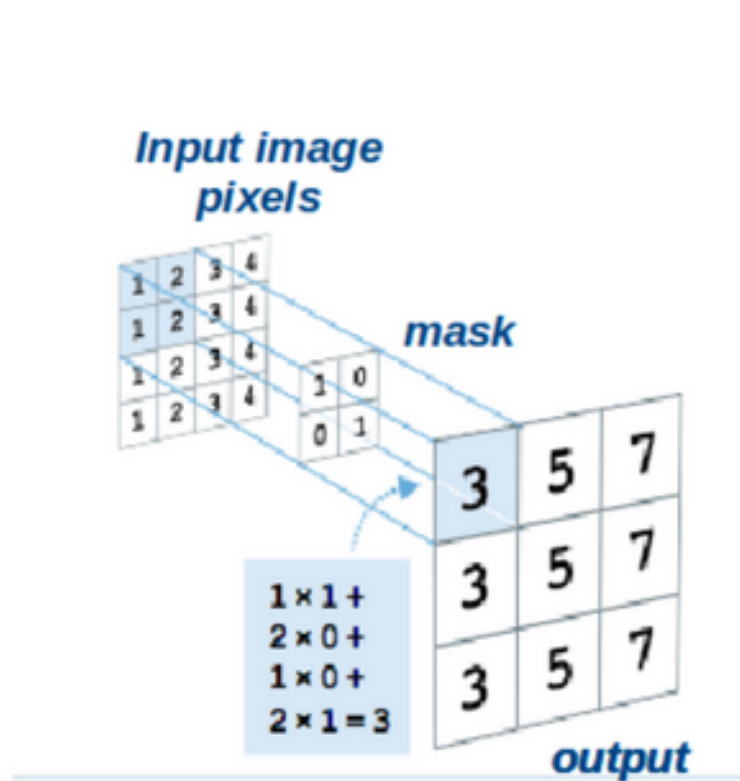


Dissecting a NN architecture

- *Flagel et al. 2019* network



How?



Convolution operation

- Convolution with padding and stride=1 → no dimension reduction

0	0	0	0	0	0	0
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0	73	121	54	84	128	0
0	131	99	70	129	127	0
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Kernel

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0	-1	0

114	328	-26	470	158
53	266	-61	-30	344
403	116	-47	295	244
108	-135	256	-128	344
314	346			

- No padding and/or stride > 1 → dimension reduction

Kernel

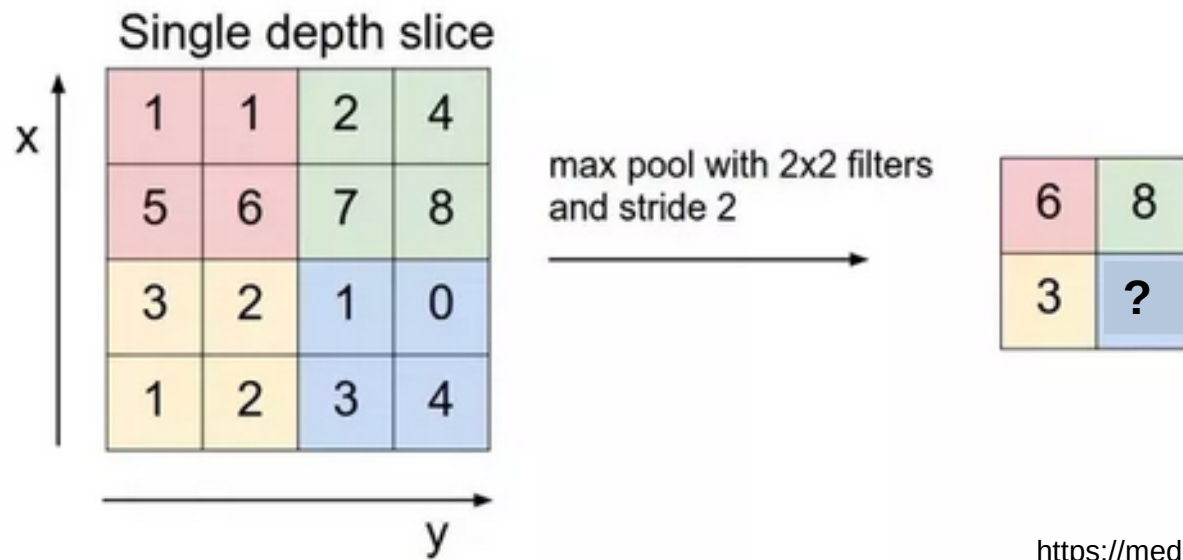
0	-1	0
-1	5	-1
0	-1	0

60	113	56	139	85
73	121	54	84	128
131	99	70	129	127
80	57	115	69	134
104	126	123	95	130

266	-61	-30
116	-47	295
-135	256	-128

Pooling operation

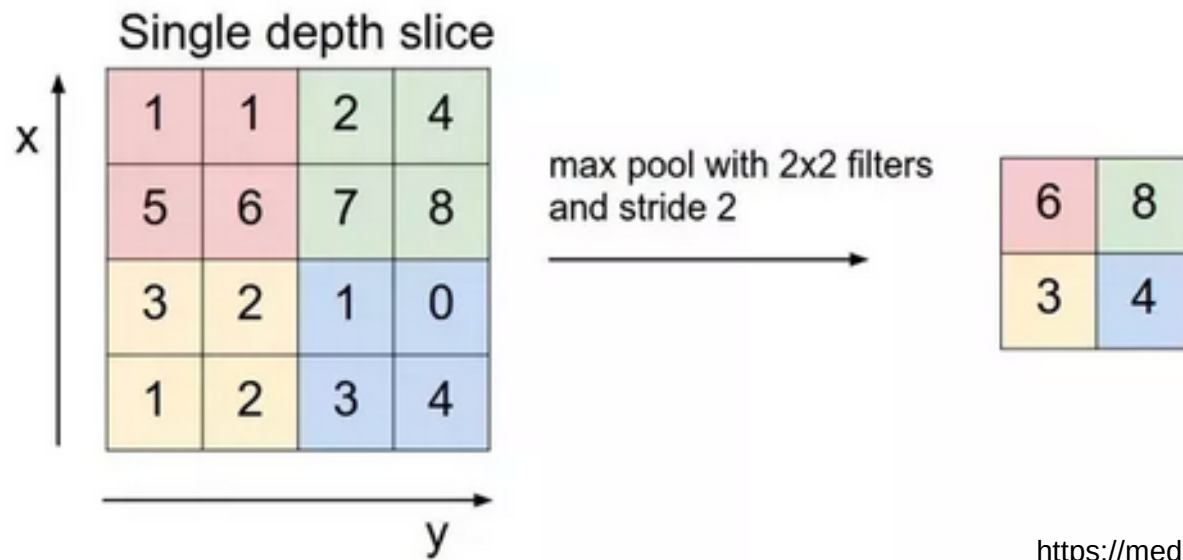
- Pooling (max, average, sum pooling)
-> reducing dimension without additional parameter to learn



<https://medium.com/@RaghavPrabhu>

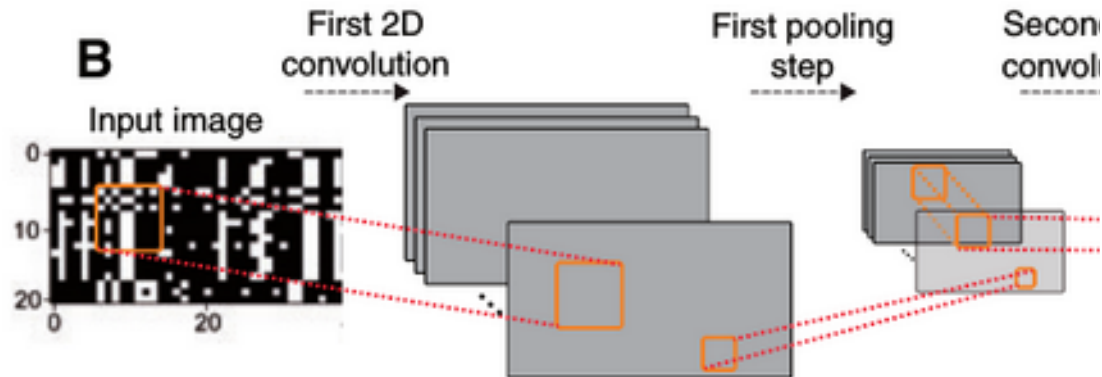
Pooling operation

- Pooling (max, average, sum pooling)
-> reducing dimension without additional parameter to learn



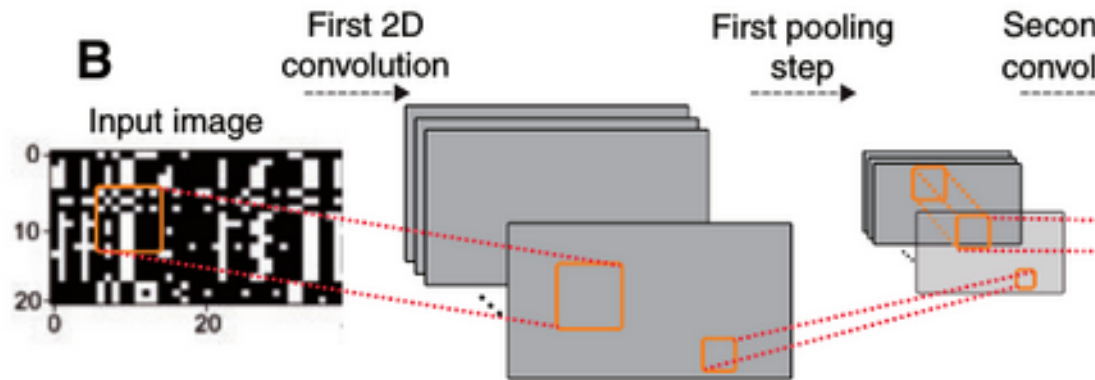
<https://medium.com/@RaghavPrabhu>

Why a 2D convolution ?

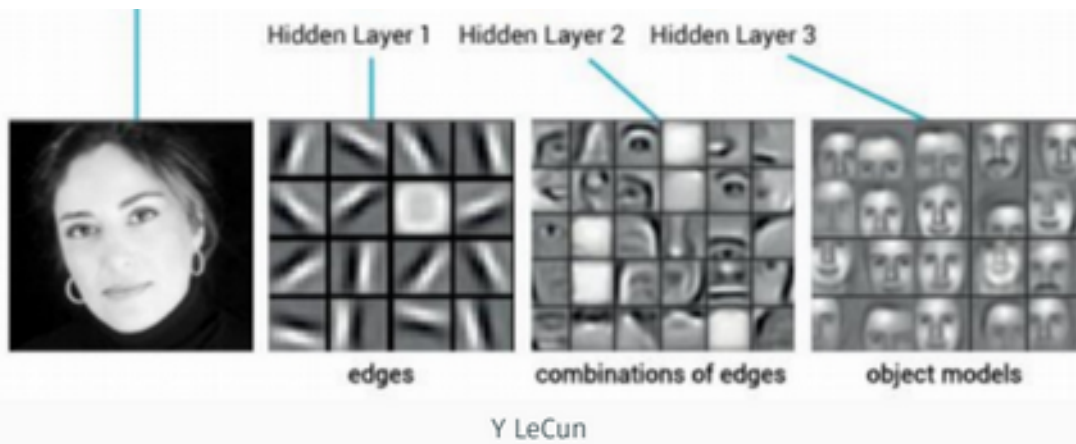


Flagel et al. 2019

Why a 2D convolution ?

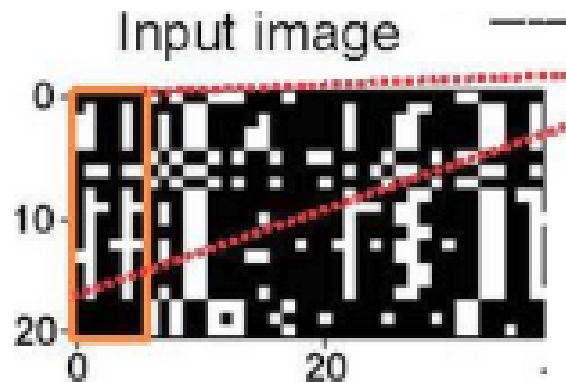


Flagel et al. 2019



But popgen data is different from a classical image!

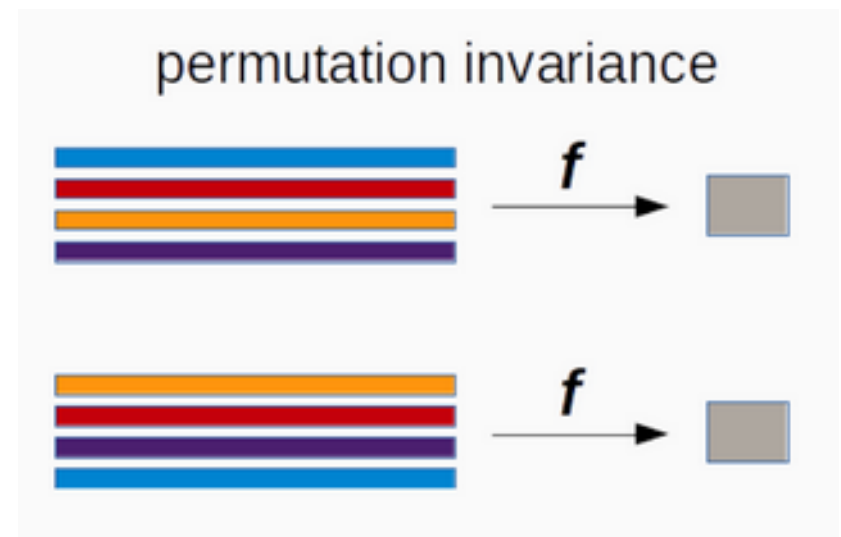
Specificity of the input genetic data



The predicted history should be invariant to the order of the individuals in the matrix

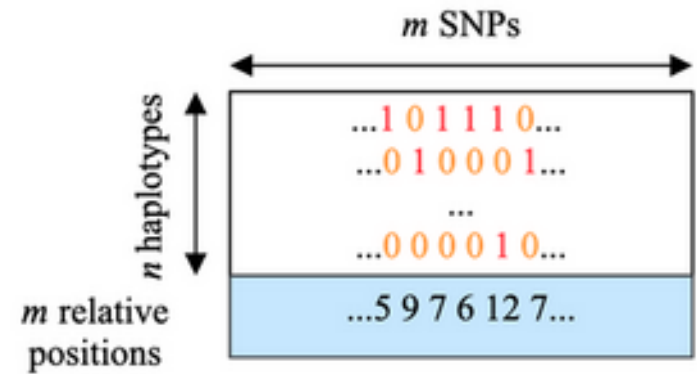
Genetic dataset D

- Ind. 1 ...ACGTTAGTGATTTTGA...
- Ind. 2 ...ACTTTACTGATTATGA...
- ...
- Ind. 3 ...ACTTTAGTGATTATGA...



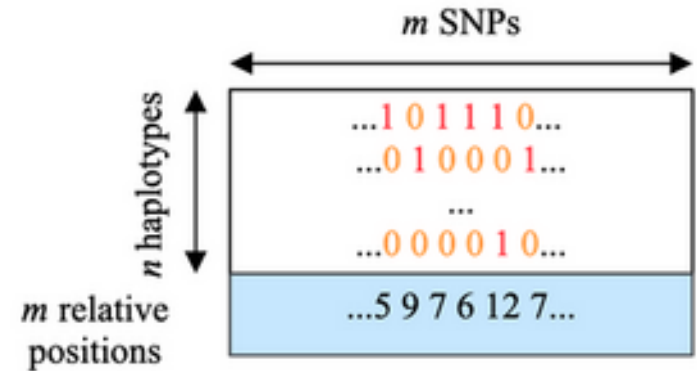
How to be invariant ?

- Any idea?



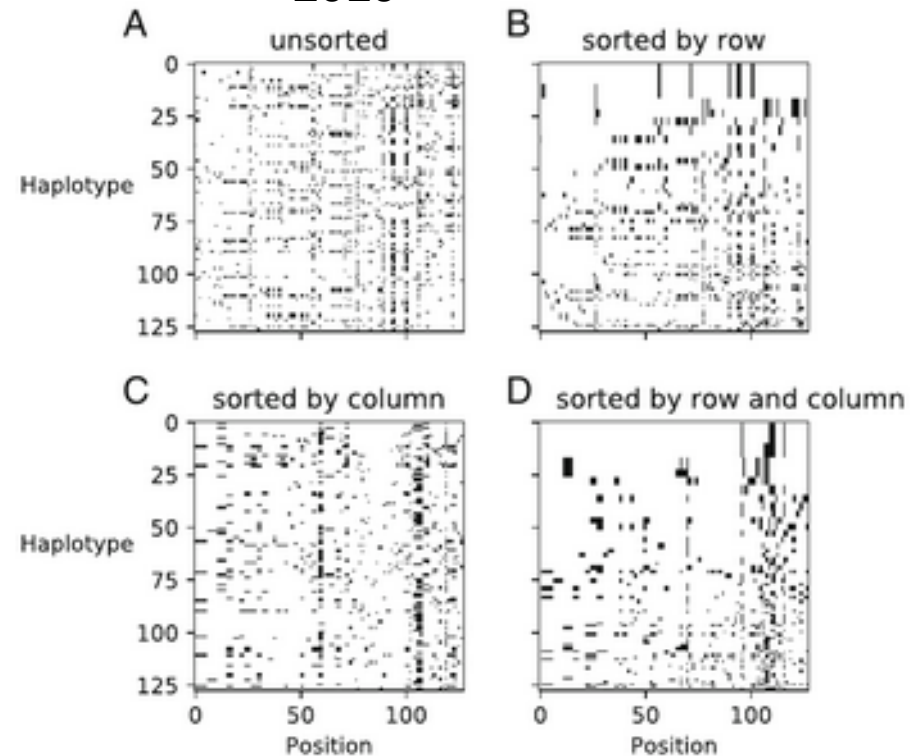
How to be invariant ?

- Solution 1: specify an **order**
ex: individual similarity (rows)
SNP similarity (columns)



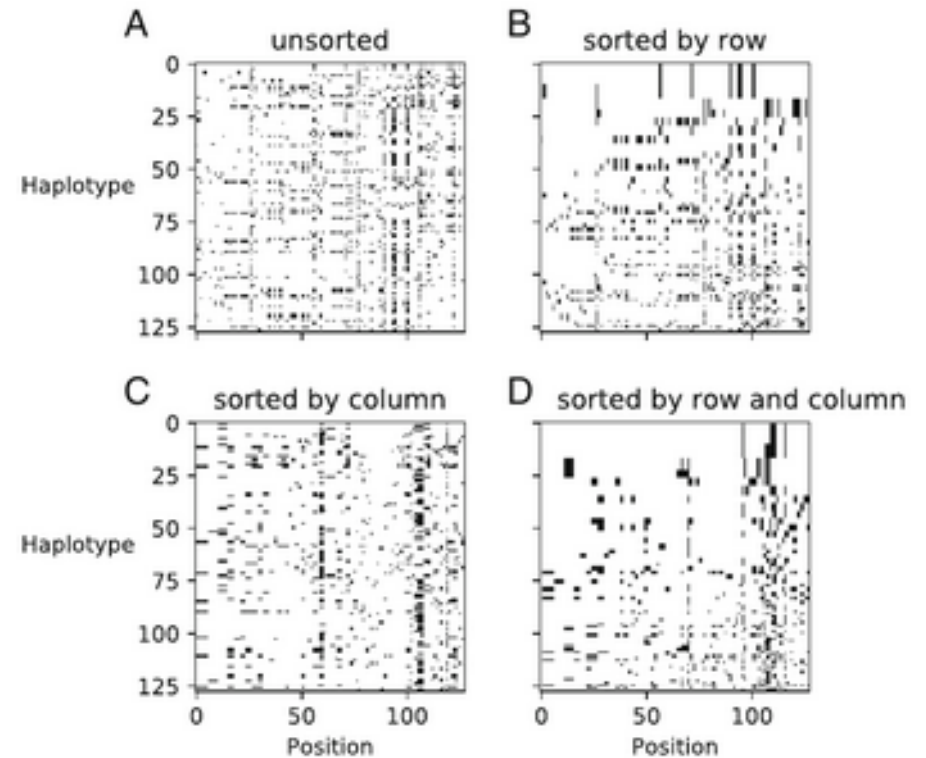
Torada et al.
2019

Flagel et al. 2018

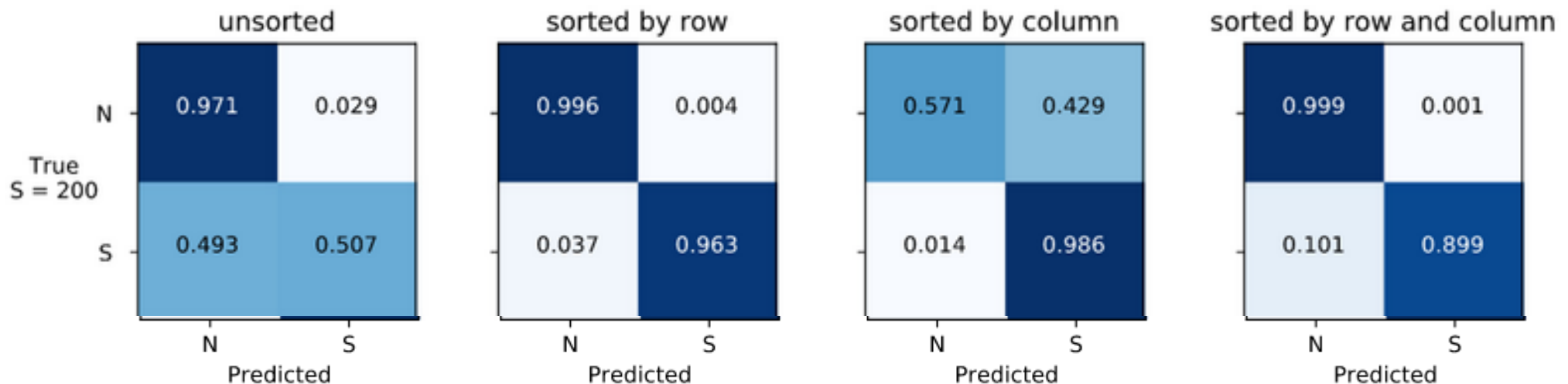


How to be invariant ?

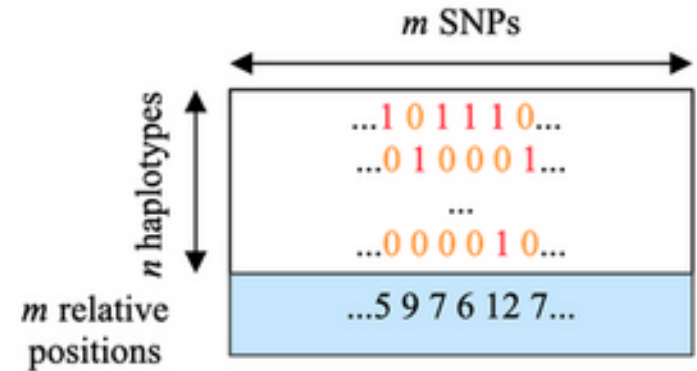
- Solution 1: specify an **order**
ex: individual similarity (rows)
SNP similarity (columns)



Torada et al. 2019
(neutral or under positive selection?)



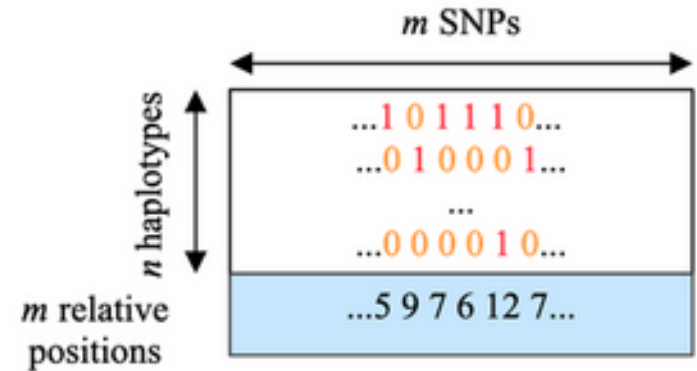
How to be invariant ?



- Solution 1: specify an **order** (ex: individual similarity) (e.g. Flagel et al. 2018, Torada et al. 2019, ...)
- Solution 2: data **augmentation**: apply transformation(s) to the input data that should not affect its label.
Ex. in some tasks of computer vision: rotating images

Exercise: give examples of data augmentation relevant for population genomics

How to be invariant ?

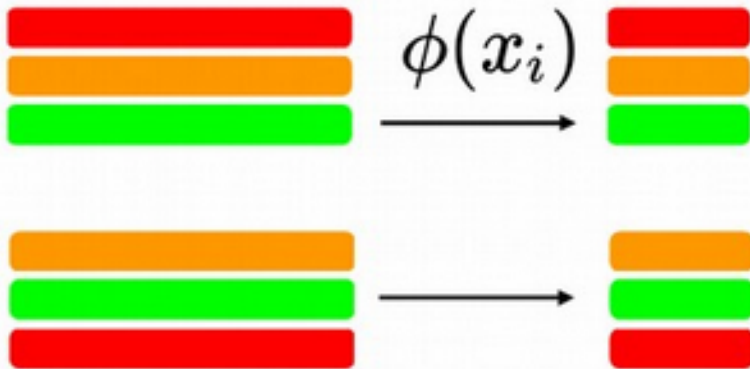


- Solution 1: specify an **order** (ex: individual similarity) (e.g. Fligel et al. 2018, Torada et al. 2019, ...)
- Solution 2: data **augmentation** (e.g. shuffling the lines)
- Solution 3: **encode invariance** in the network (permutation-invariant network, exchangeable network) (e.g. Chan et al. 2018, Wiqvist et al. 2019, Sanchez et al 2020, Sanchez et al 2026)

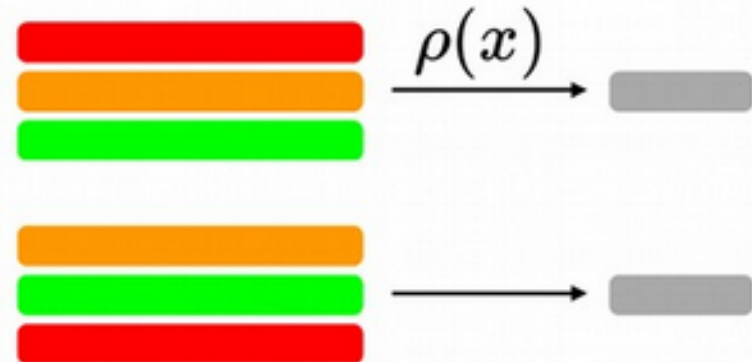
Adapt the network to the data features

- **Invariant** to the permutation of rows

Equivariant function



Invariant function



Exercise: give an example of invariant operation

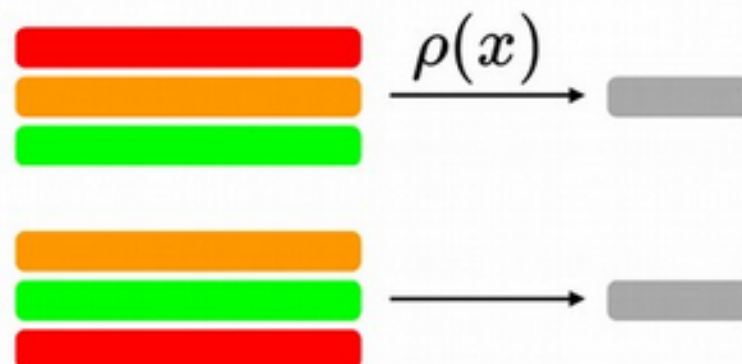
Adapt the network to the data features

- **Invariant** to the permutation of rows

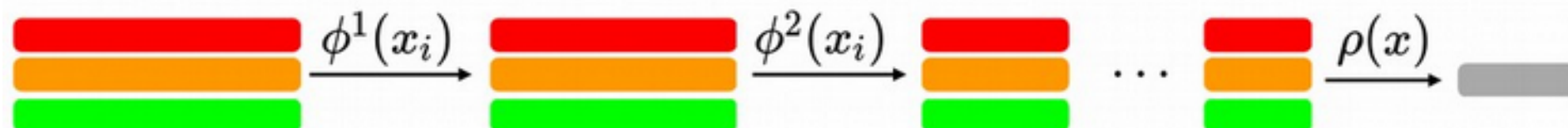
Equivariant function



Invariant function



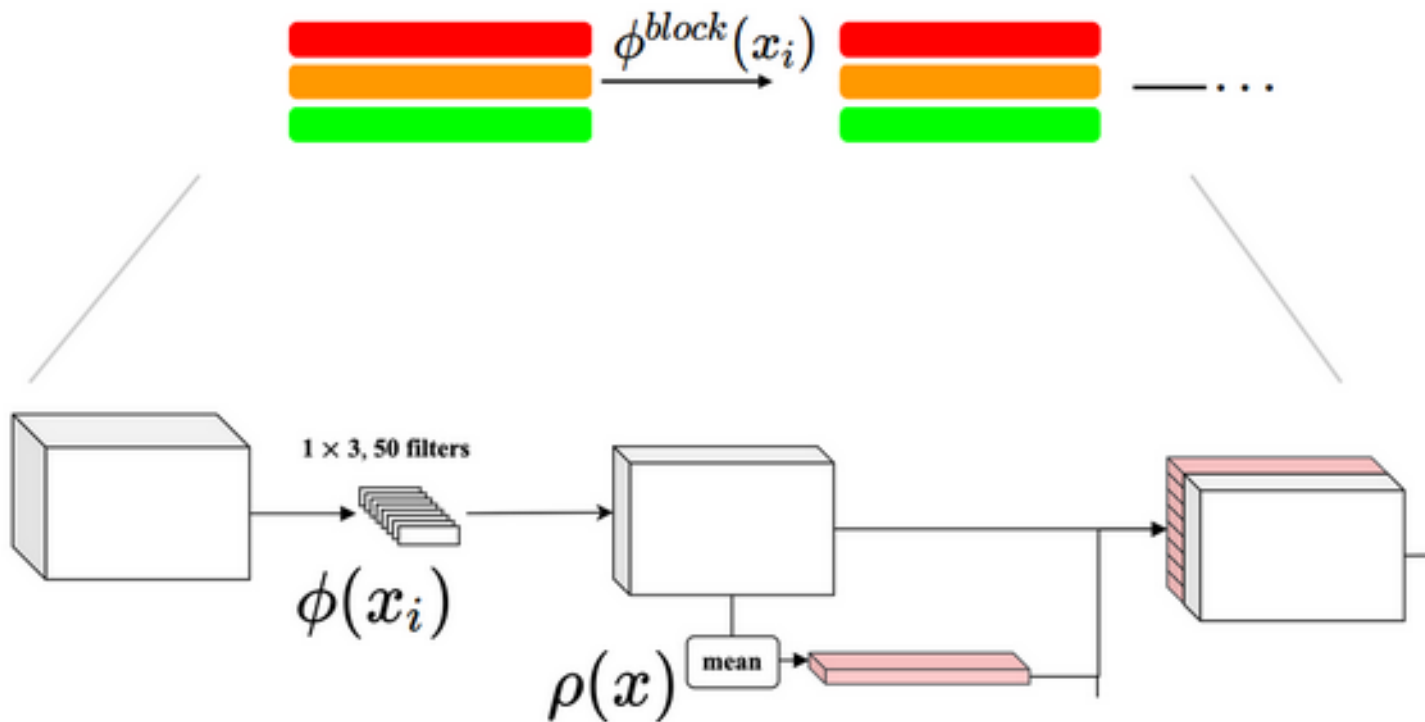
Stacking invariant and equivariant layers **covers the full space of permutation invariant** function (Zaheer et al. 2017, Lucas et al. 2018)



Adapt the network to the data features

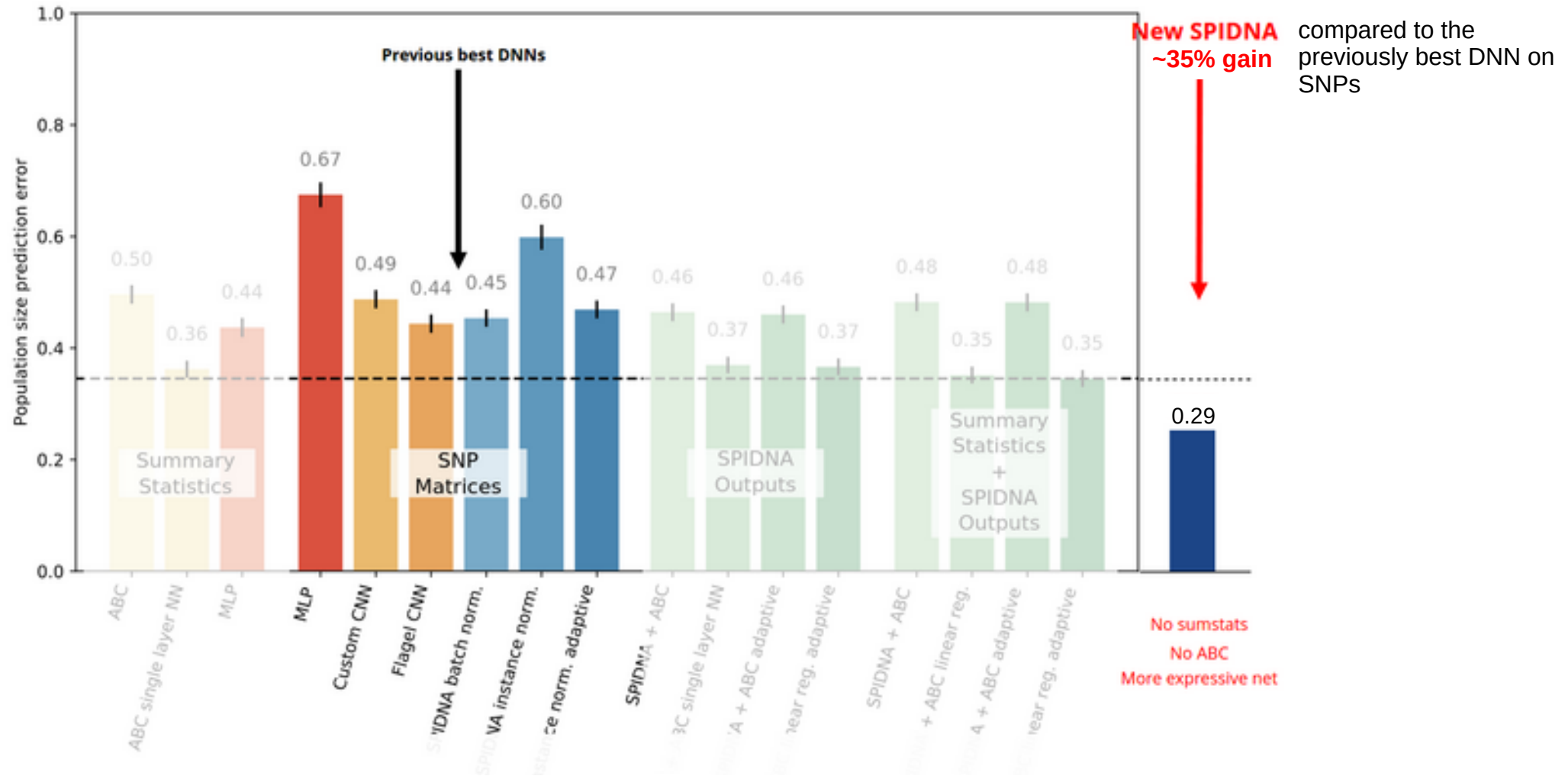
- Each block of SPIDNA defines an equivariant function

Sanchez et al 2020



Sanchez et al 2026 further extends the expressivity of this equivariant function using attention mechanisms <https://www.biorxiv.org/content/10.64898/2026.04.15.718687v1>

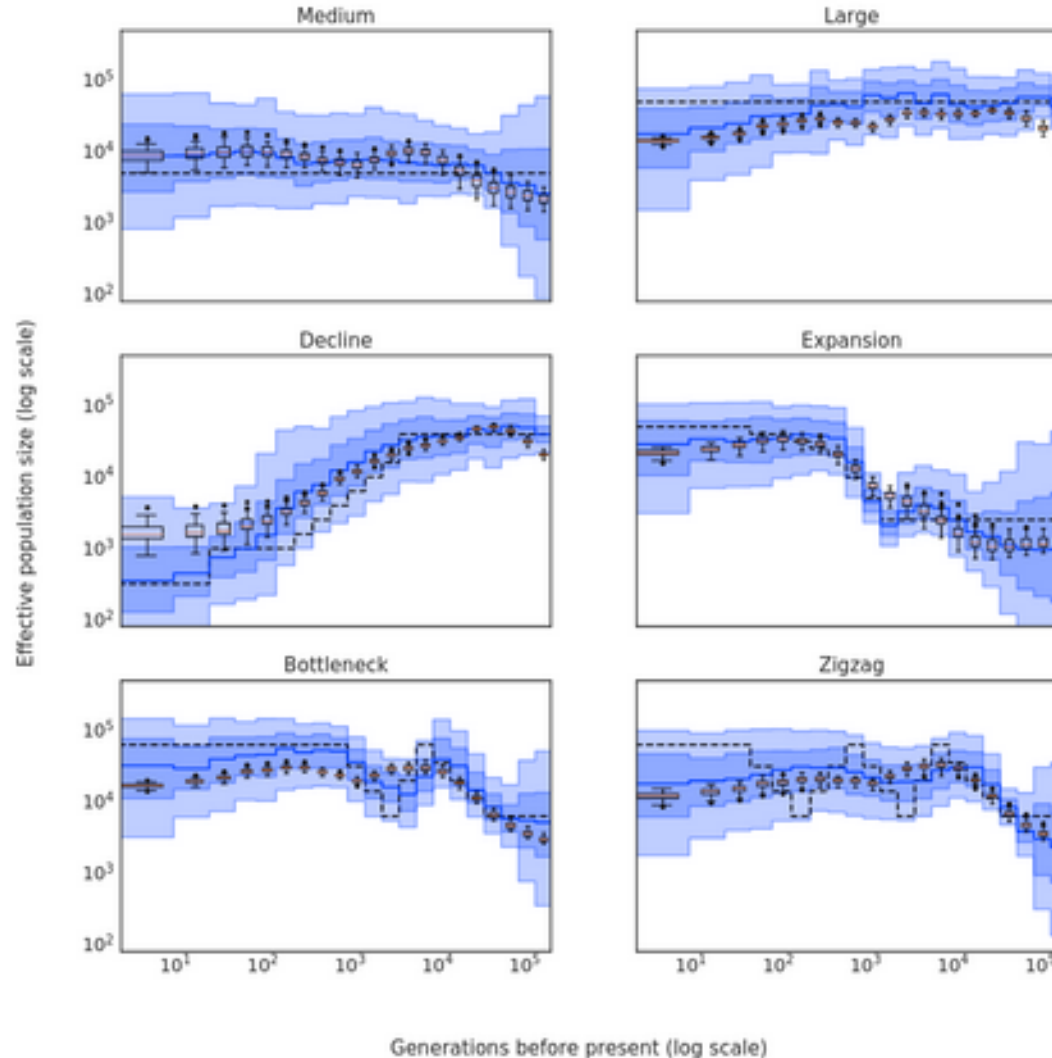
Comparison of prediction error on simulated data



(with attention for sequences
and for genomic regions)

Predictions for handmade scenarios

(scenarios from popsizeABC, Boitard et al. 2016)



Posterior approximated with
Approximate Bayesian Computation
 $P(\theta|X) = P(X|\theta)\pi(\theta)/P(X) \propto P(s(X)|\theta)\pi(\theta)$

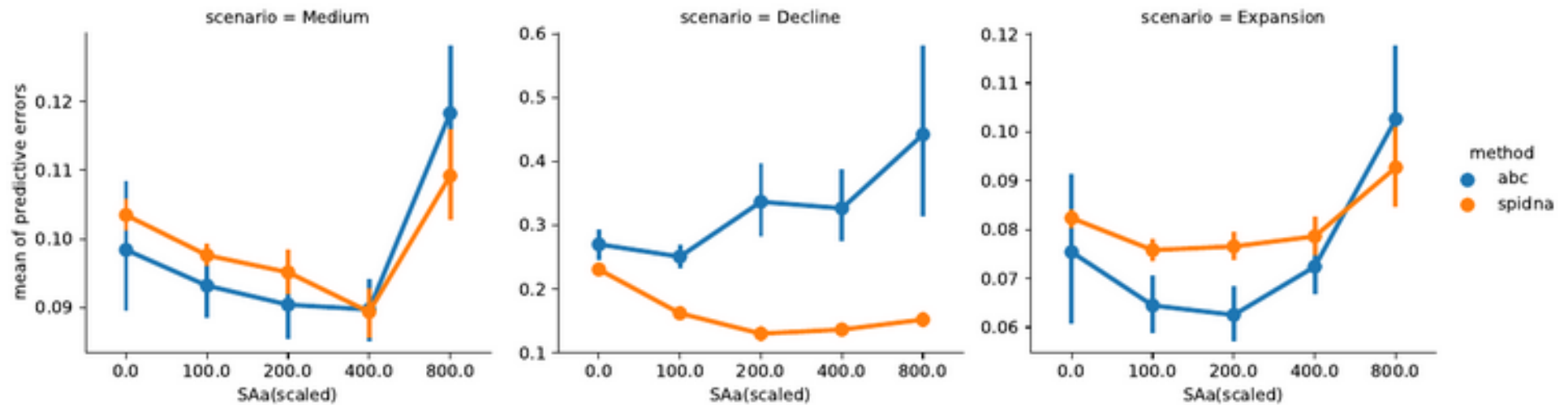
θ : evolutionary parameters

X : observed genetic data

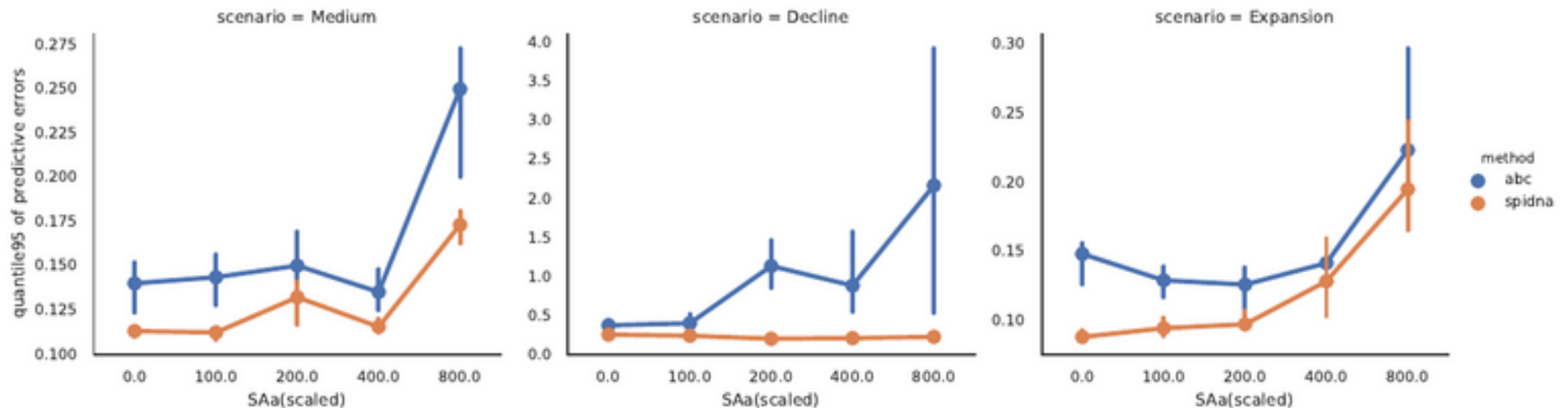
$s(.)$ function mapping to (automatic or not) sumstats

SPIDNA can be more robust than **ABC** to model misspecification (presence of positive selection)

Mean predicted error on subset of scenarios

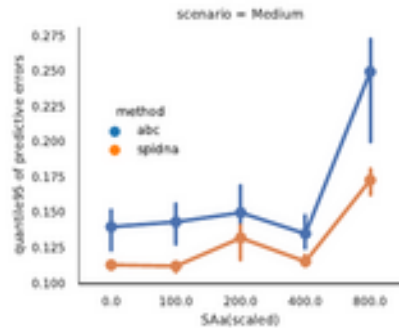


95% quantile of predicted error



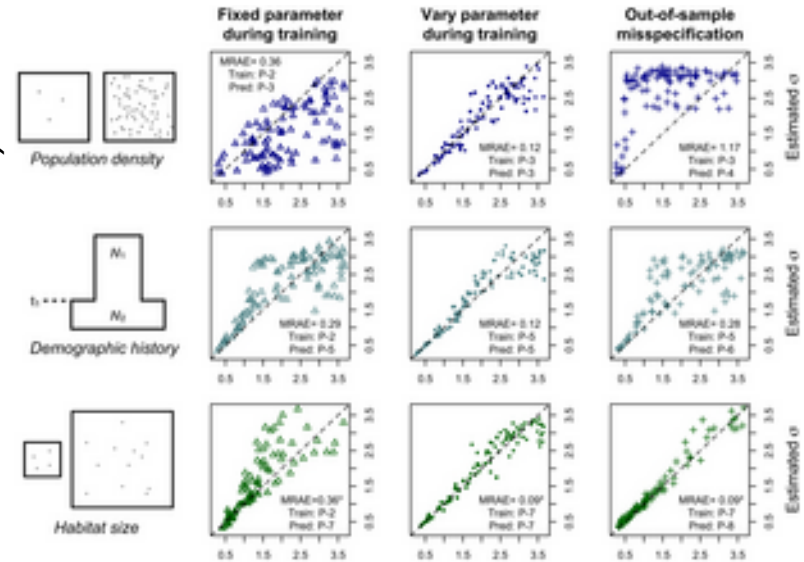
Robustness and uncertainty

- Robustness to misspecifications frequently tested in the field



Sanchez et al 2021
(unmodeled selection)

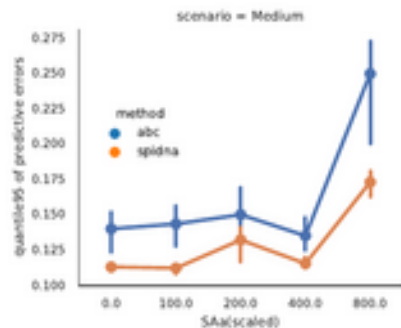
Smith et al 2023 (dispersal
inference with CNN)



Robustness and uncertainty

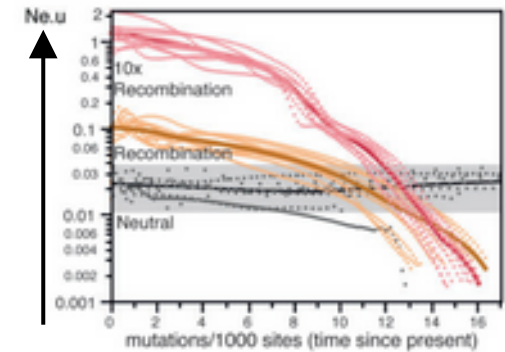
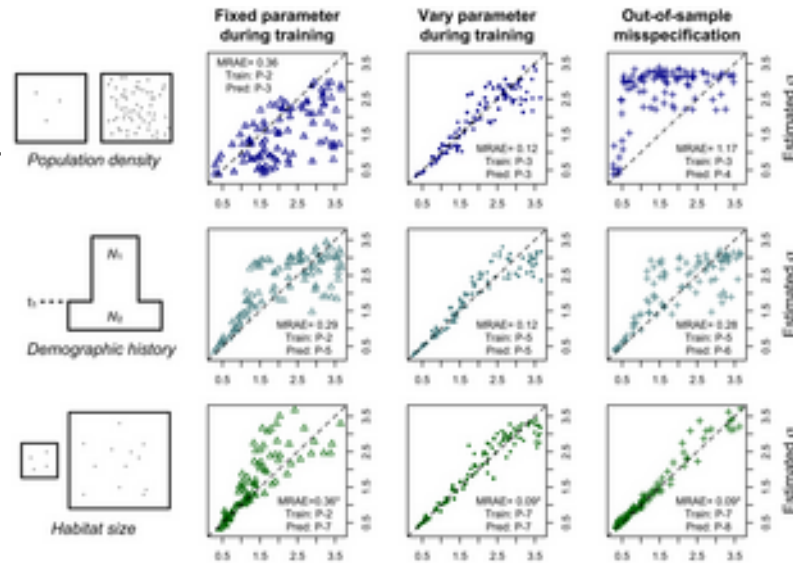
- Robustness to misspecifications frequently tested in the field

- Not unique to DNNs



Sanchez et al 2021
(unmodeled selection)

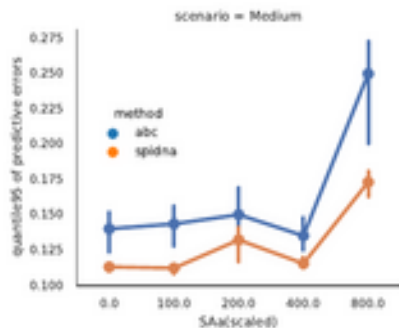
Smith et al 2023 (dispersal
inference with CNN)



Lapierre et al. 2016
(testing robustness of skyline plot)

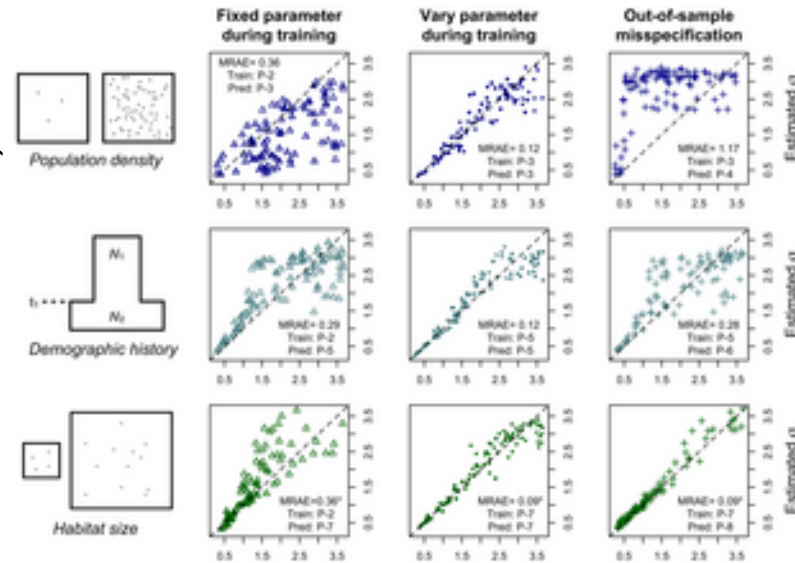
Robustness and uncertainty

- Robustness to misspecifications frequently tested in the field

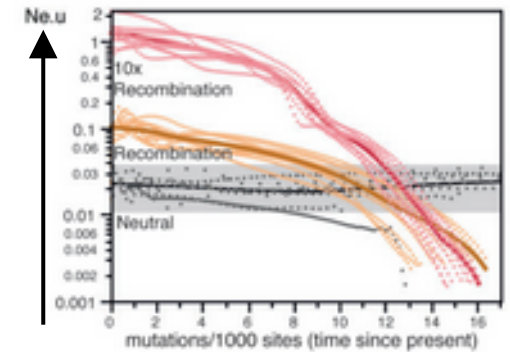


Sanchez et al 2021
(unmodeled selection)

Smith et al 2023 (dispersal inference with CNN)

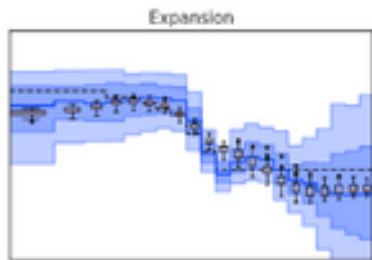


- Not unique to DNNs

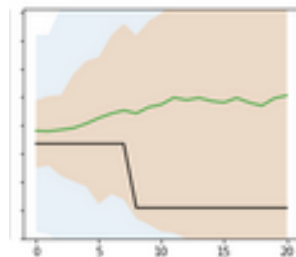


Lapierre et al. 2016
(testing robustness of skyline plot)

- Uncertainty can be inferred for DNNs



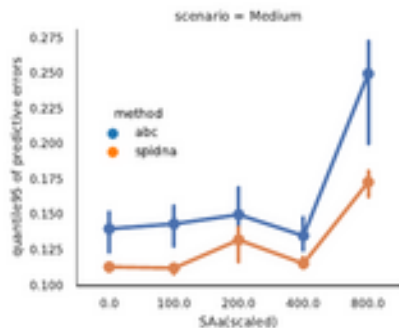
Sanchez et al 2021



Cury et al 2022

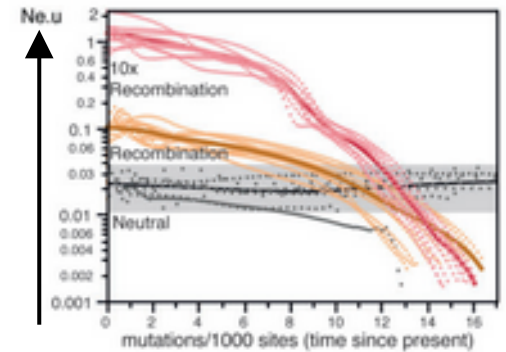
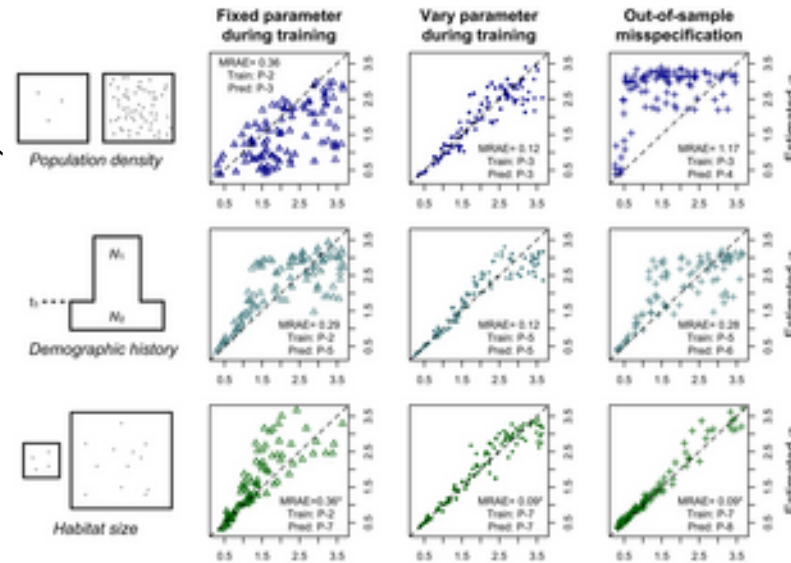
Robustness and uncertainty

- Robustness to misspecifications frequently tested in the field



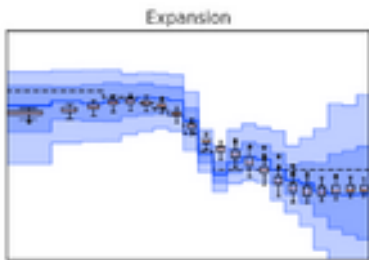
Sanchez et al 2021
(unmodeled selection)

Smith et al 2023 (dispersal inference with CNN)

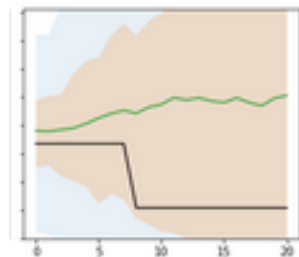


Lapierre et al. 2016
(testing robustness of skyline plot)

- Uncertainty can be inferred for DNNs



Sanchez et al 2021



Cury et al 2022

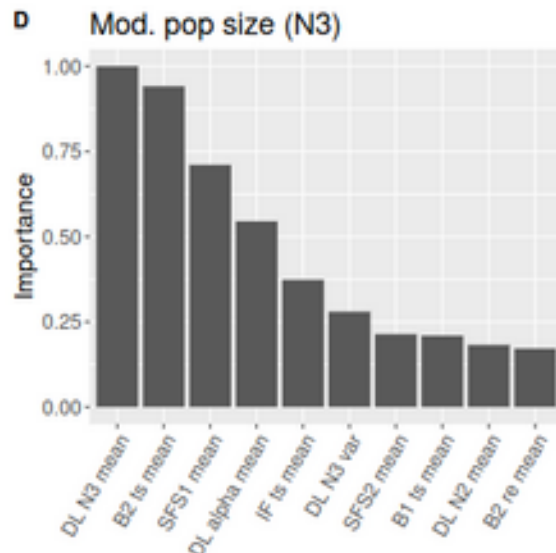
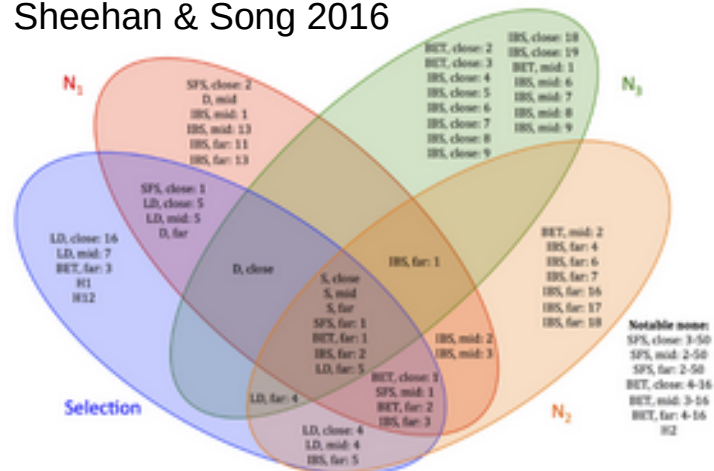
- Domain adaptation could be a promising technique ; recently applied to population genetics (Mo&Siepel PloS Genet 2023)

→ learning features invariant to the domain (training or target)

Interpretability

DL on sumstats

Sheehan & Song 2016

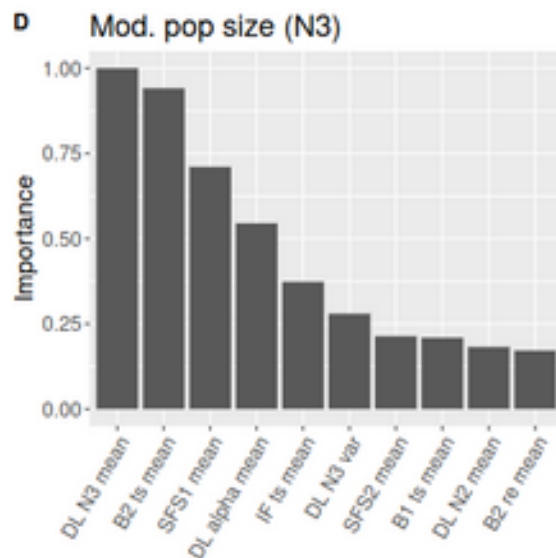
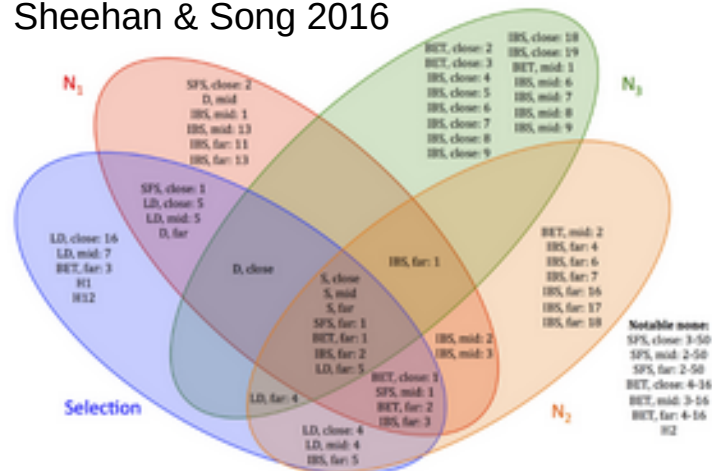


Guez 2023 "Cultural transmission of reproductive success" thesis
 see also Burger et al 2022; Quelin et al 2025

Interpretability

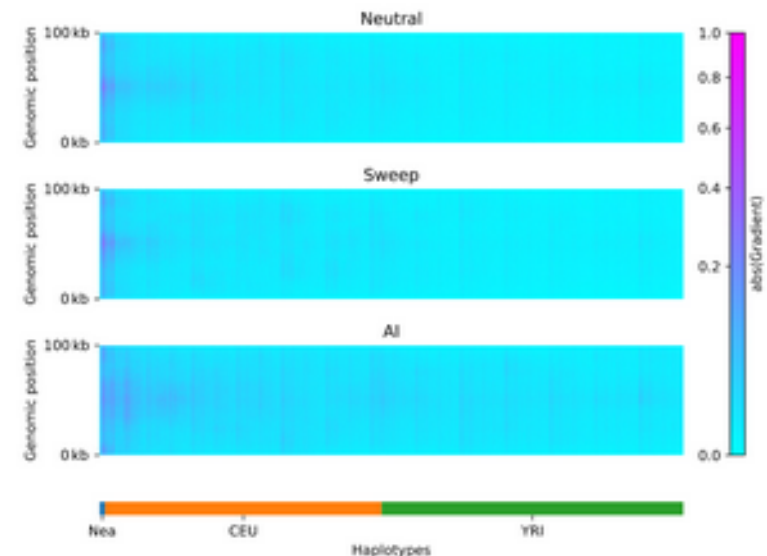
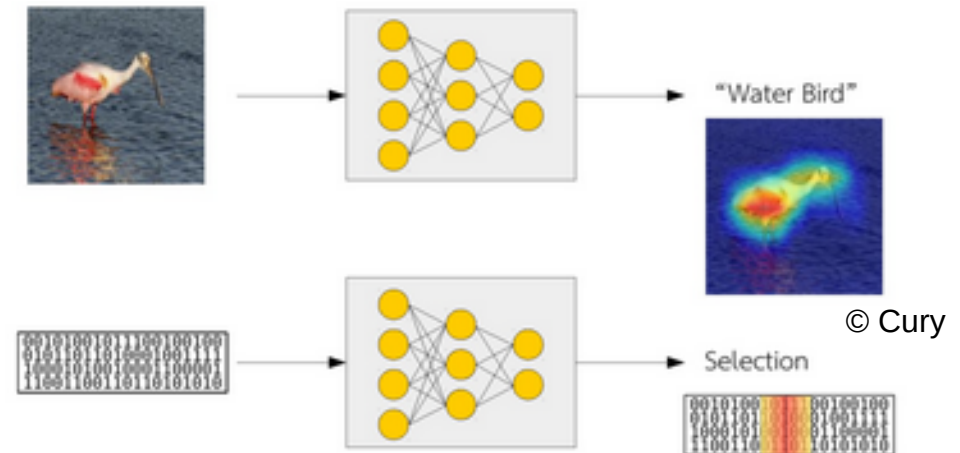
DL on sumstats

Sheehan & Song 2016



Guez 2023 “Cultural transmission of reproductive success” thesis
see also Burger et al 2022; Quelin et al 2025

DL on raw data



Gower et al 2021 (detecting adaptive introgression):
averaged saliency maps (Simonyan et al., 2014)
see also Yelmen et al 2026 (for GWAS)

Final note on DL applied to population genetics

- Still a **recent** “combo” (2016 for DL on summary statistics, 2018 for DL on raw population genetic data) -> space for creativity and new proposals!
- Remember some important rules in stat/ML/DL:
 - **Think** properly about your **task**, statistical **model**, **evaluation** scheme/metrics
You need proper null models
 - Watch out for **overfitting**, use train/validation sets
 - **Compare** to previously published methods
 - Choose or explore **hyperparameter space** for each approach (grid search, Bayesian hyperoptimisation, ...) based on validation set
 - Final comparison on an **independent test set**

Deep learning hyper-parameters (HP)

- You still have to make decisions about (1) your architecture (#layers, #nodes per layer, layer type,...) ; (2) the algorithm/optimization hyper-parameters
- Usually done by training numerous networks with numerous HP and keeping the one performing the best. Can be done in a smart way with e.g. bayesian HP optimization. Automatic Deep Learning : active area of research

**Just a taste of
some NN algo
hyper-parameters**

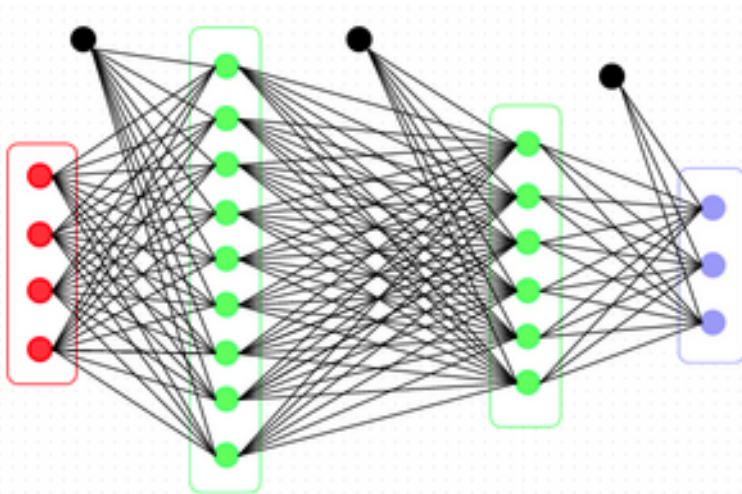


Table 2. Central parameters of a neural network and recommended settings.

Name	Range	Default value
Learning rate	0.1, 0.01, 0.001, 0.0001	0.01
Batch size	64, 128, 256	128
Momentum rate	0.8, 0.9, 0.95	0.9
Weight initialization	Normal, Uniform, Glorot uniform	Glorot uniform
Per-parameter adaptive learning rate methods	RMSprop, Adagrad, Adadelta, Adam	Adam
Batch normalization	Yes, no	Yes
Learning rate decay	None, linear, exponential	Linear (rate 0.5)
Activation function	Sigmoid, Tanh, ReLU, Softmax	ReLU
Dropout rate	0.1, 0.25, 0.5, 0.75	0.5
L1, L2 regularization	0, 0.01, 0.001	

Final note on DL applied to population genetics

- Still a **recent** “combo” (2016 for DL on summary statistics, 2018 for DL on raw population genetic data) -> space for creativity and new proposals!
- Remember some important rules in stat/ML/DL:
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 - **Compare** to previously published methods
 - Choose or explore **hyperparameter space** for each approach (grid search, Bayesian hyperoptimisation, ...) based on validation set
 - Final comparison on an **independent test set**
- Evaluate method **robustness** to data **corruption**, model **misspecification**, ...
Particularly relevant for simulation-based inference approaches where simulators and simulation scenarios have underlying **assumptions** that can be violated in the real life.
- THINK CAREFULLY! You might gain in accuracy but lose in **uncertainty estimation** and **explainability**/interpretability (ex: CNN versus a previous approach based on SFS). Improving uncertainty estimation/interpretability is actively studied in DL field.

Outline

Machine Learning: basic concepts and terminology

ML, application to popgen ; neural networks

- I. From ABC to deep learning for population genetics
- II. Learning directly from SNP data with neural networks
- III. Dissecting two published networks for effective population size inference
- IV. Opening on applications of unsupervised deep learning to popgen (if you want to have a look at home)**
- V. Hands-on: building/training/re-using ML and DL models with application to population genetics (demography)**

Announcements

- **LEGEND conference on machine learning for evolutionary genomics**, 3rd edition should be in Germany in **2027** (see previous edition: <https://legend2025.sciencesconf.org/>)
- In France: LEGO working group <https://gt-lego.cnrs.fr/>

Tenure track position in AI for genomics/evolutionary genomics opening in my lab (LISN, Paris-Saclay) or in LBBE (Lyon), France. **Please advertise in your groups!** (people can contact me at flora.jay@lisn.fr for any question)

<https://www.ins2i.cnrs.fr/en/cnrsinfo/list-cnrs-informatics-tenure-tracks-2026>



Unsupervised / Supervised Tasks

- Learning something from **data**
- Either **unsupervised** (no labels) or **supervised** (discrete or continuous labels)

Unsupervised learning

- **Learning something from data without labels**
- **Unsupervised = discovering patterns in data without prior knowledge**

You do NOT have labels, or you do NOT use them

Exercise: Give examples of unsupervised tasks in population genetics

-
-
-

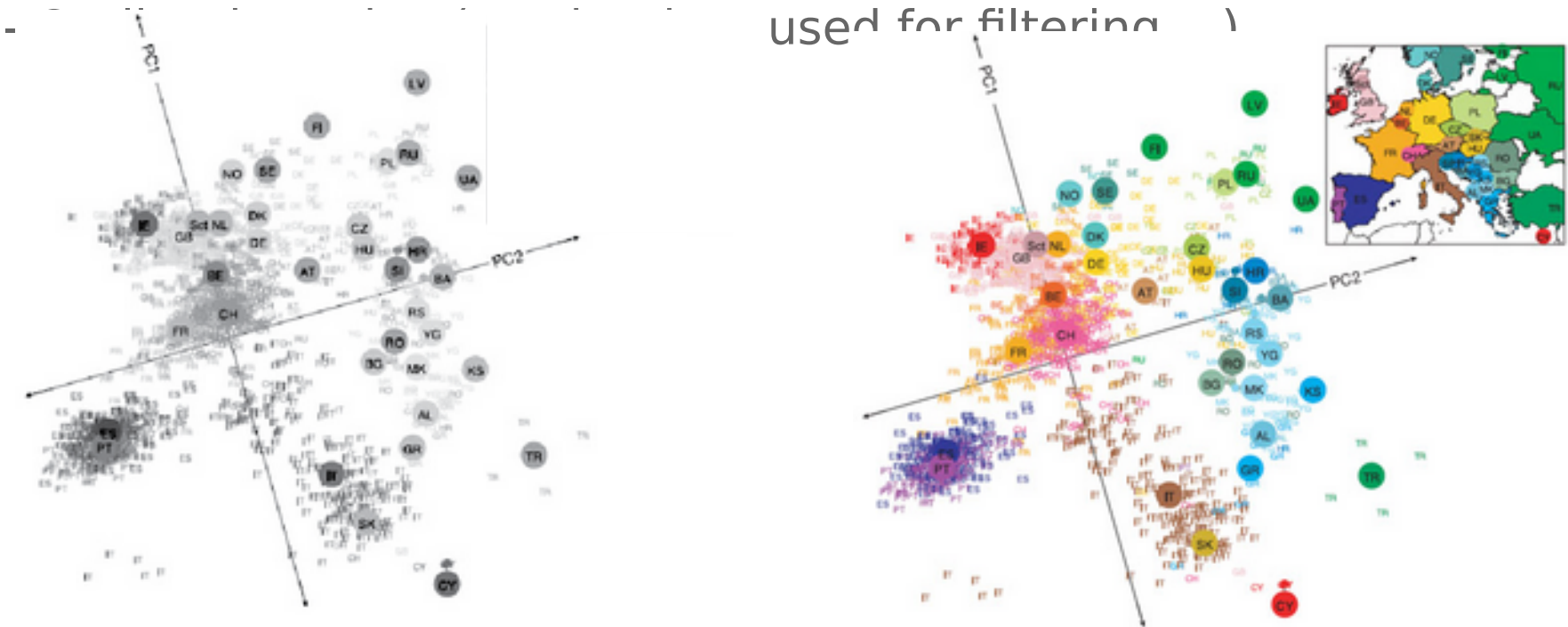
Unsupervised learning

- **Unsupervised = discovering patterns in data without prior knowledge**
You do NOT have labels, or you do NOT use them
 - Dimension reduction methods, e.g. PCA, Matrix factorization
 - Clustering algorithms, e.g. K-means, hierarchical clustering, ...
 - Outlier detection (can be then used for filtering, ..)
 - ...

Unsupervised learning examples (not only deep neural nets here)

- Dimension reduction methods, e.g. PCA, Matrix factorization
- Clustering algorithms, e.g. K-means, hierarchical clustering, SNMF

...



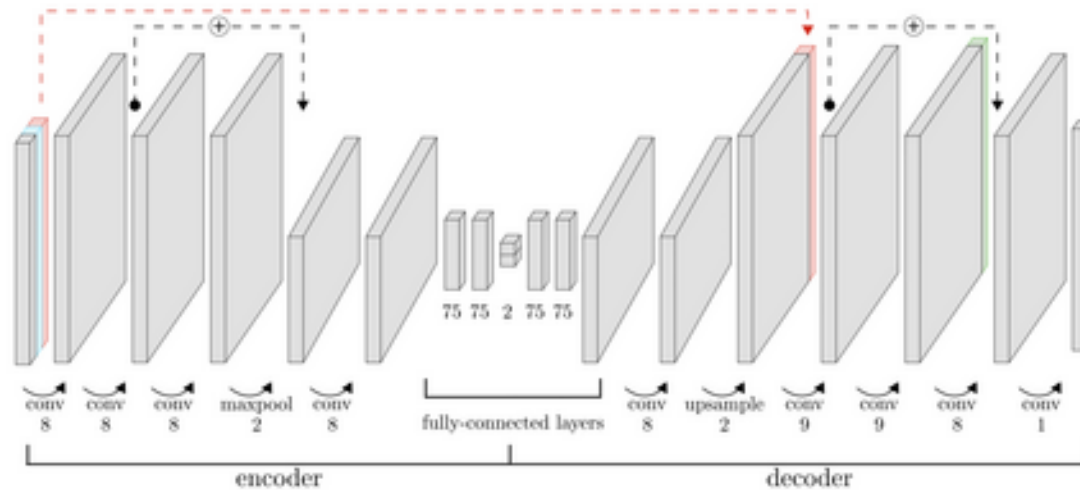
PCA to reduce high dimensional genotype data for human populations

The 1st axis (ie the linear combination of markers) explains the largest part of the variance among samples. The 2nd axis explains the largest part of the remaining variance, and so on..

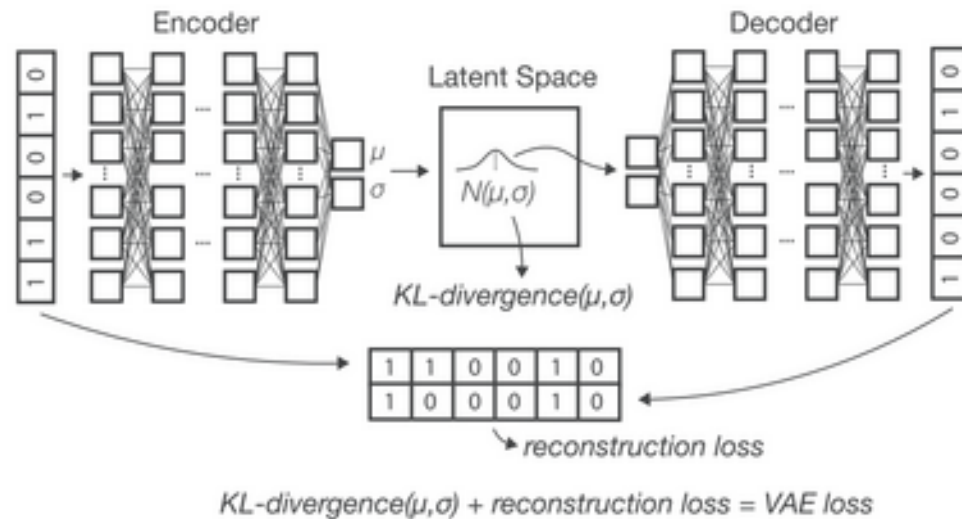
Novembre et al 2008

Neural networks for unsupervised tasks

(i) reconstructing oneself after a strong reduction in dimension



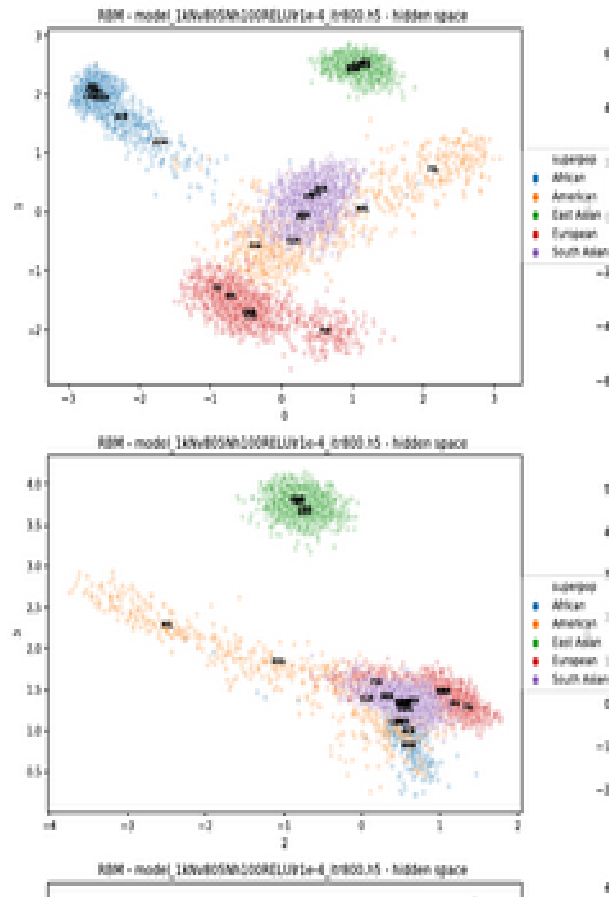
Convolutional Autoencoder (AE)
Ausmees et al 2021



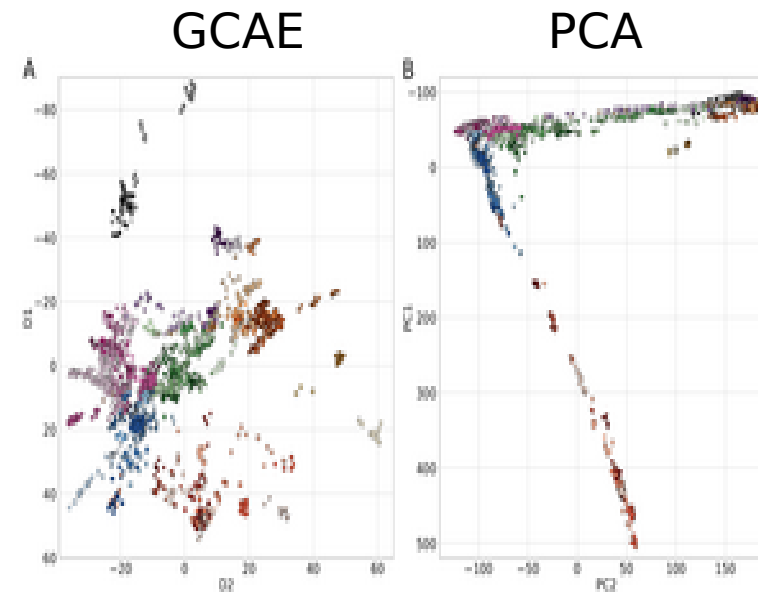
Variational Autoencoder (VAE)
Battey et al 2021

Non-linear dimension reduction based on neural networks for visualizing genetic data

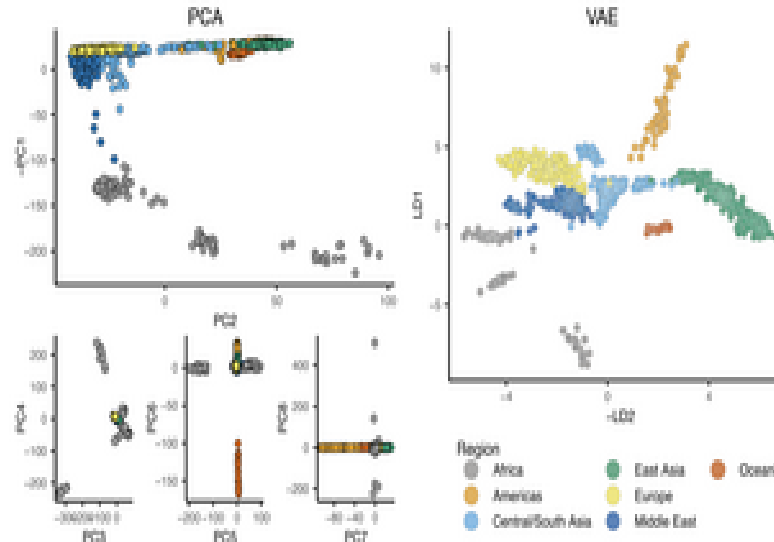
Yelman et al. 2021 ; Yelman et al. (2023)
restricted Boltzman machine ; GAN ; VAEs



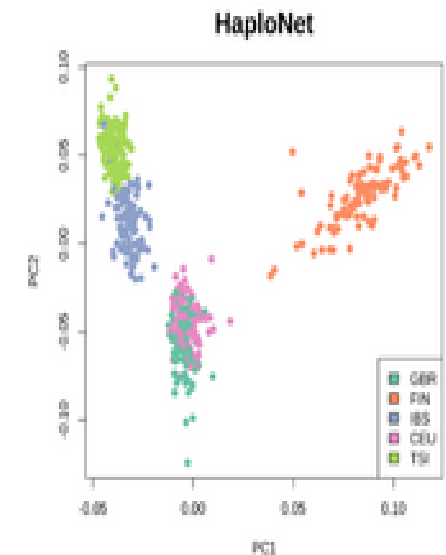
Ausmees & Nettelblad 2022 convolutional autoencoders



Battey et al. 2021
variational autoencoders (VAE)



Meisner & Albrechtsen 2022
VAE



Generative models (unsupervised learning)

Data with no label

Goal Generate samples having the same distribution as the data

Training data $\sim p_{\text{data}}(x)$
(distribution unknown)

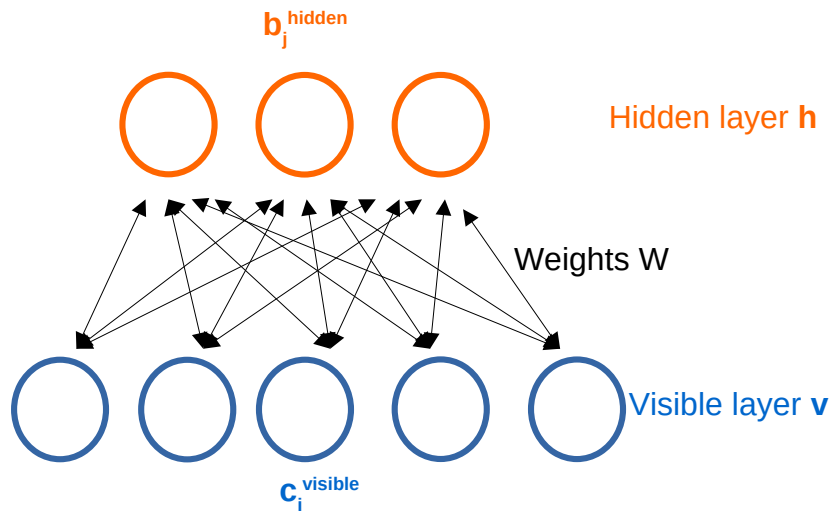


Generated samples $\sim p_{\text{model}}(x)$



Neural networks for unsupervised tasks

- (ii) generating realistic genomes that do not belong to a real individual



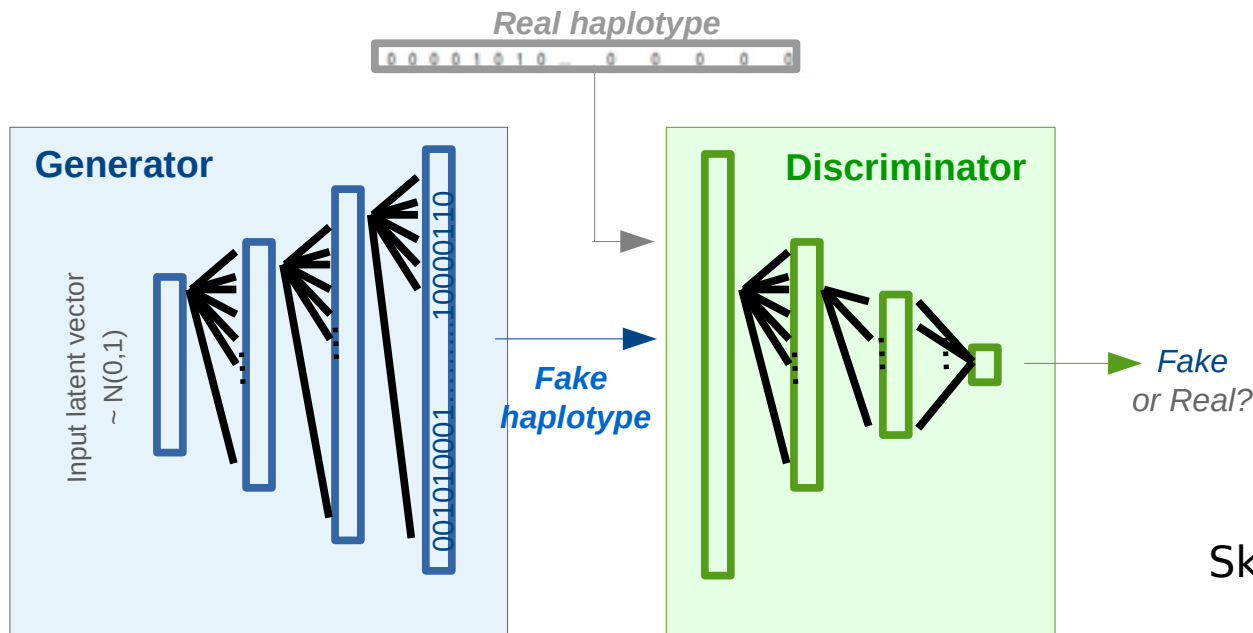
Probabilistic model of the joint distribution of $\mathbf{v}$ and $\mathbf{h}$

$$P(v, h) = e^{-E(v, h)} / Z$$

Z: partition function

$$E(v, h) = \sum_{ij} W_{ij} v_i h_j + \text{bias terms}$$

Restricted Boltzman Machine (RBM)



Generative Adversarial Networks (GAN)

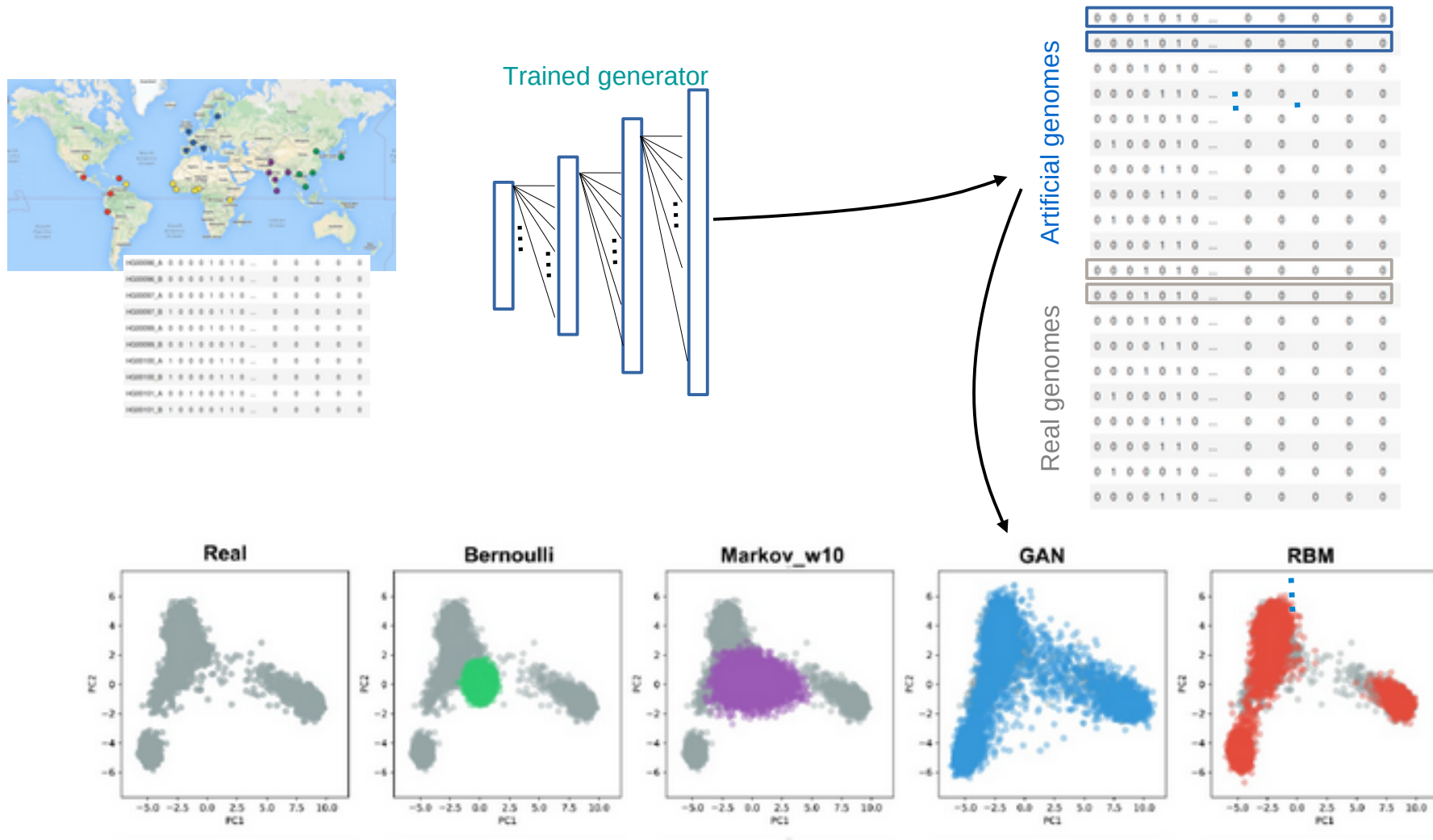
Yelmen et al 2021

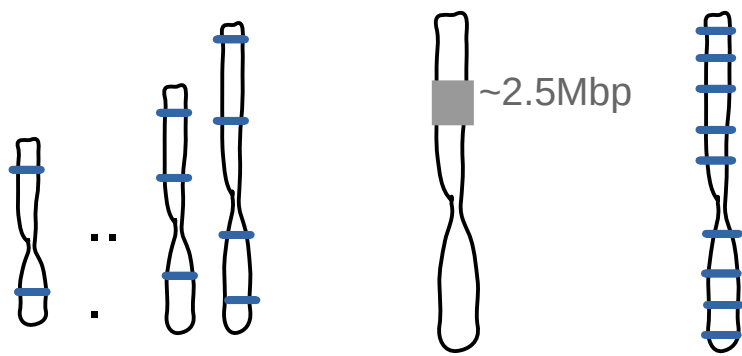
Yelmen et al 2023

Skatkownik et al (preprints on latent generation and diffusion)

Unsupervised learning for generating realistic genomes

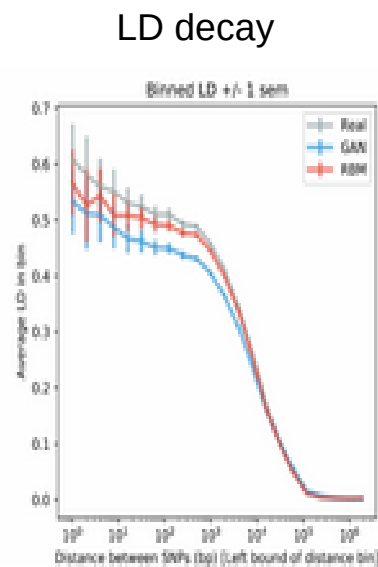
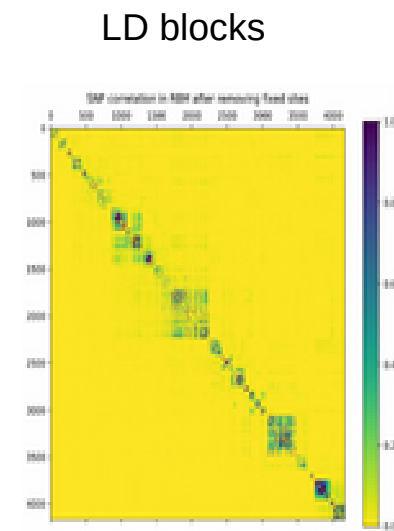
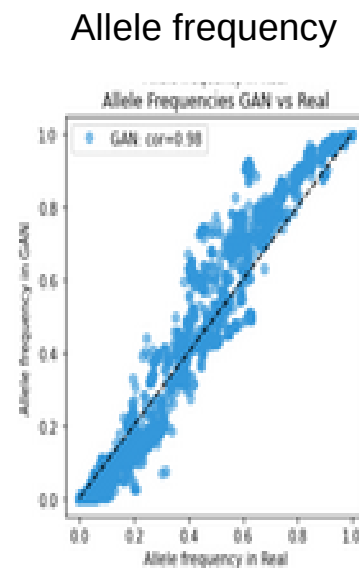
- Generative models (can also be used for dimension reduction and exploring latent space) Yelmen et al 2021/2023 (GAN, RBM, VAE neural networks) ; Battey et al 2021 (VAE) ; Ausmees et al 2021(AE)



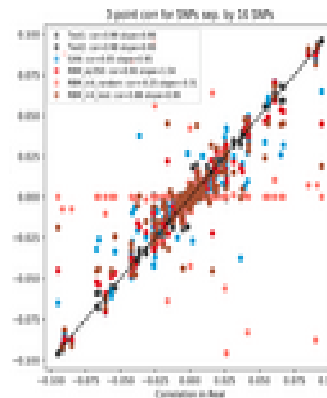


Quality checks

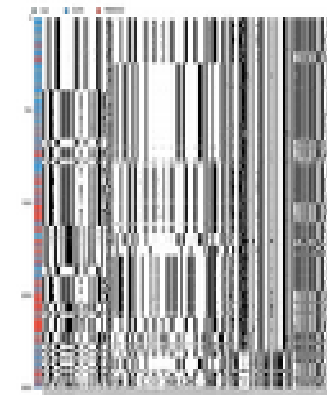
- PCA, tSNE, UMAP
- Allele frequencies (1-point correlation)
- Linkage disequilibrium patterns (2-point correlation)
- Haplotype structure (and 3-point correlation)
- Local ancestry block patterns
- Pairwise distance distributions, ...



3-SNPs correlation

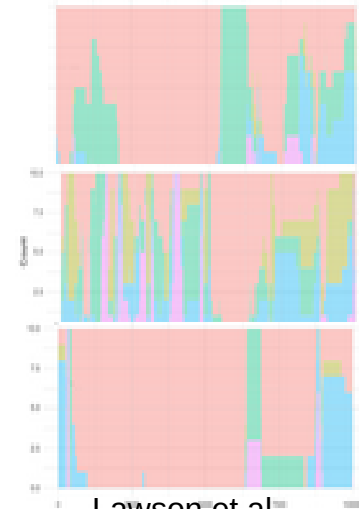


Haplotype structure



Marnetto and Huerta-Sánchez
2017

Local ancestry



Lawson et al
2012

→ Are characteristics preserved in generated genomic sequences ?